MTH321

Numerical analysis of PDEs

Mathematical formulation and methods for mixed Poisson problems

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September 16, 2026

Contents

1 Introduction **3**

2 Mathematical model **4**
2.1 Domain, boundary data and regularity 4
2.2 Manufactured benchmark 5
2.3 Weak formulation and uniqueness 6

3 Finite difference discretisation **7**
3.1 Grid, unknowns and norms 7
3.2 Interior stencil and first-order closure 7
3.3 Ghost-point closure 8
3.4 Dirichlet elimination and matrix scaling 8
3.5 Algorithm and implementation 9

4 Stability and error analysis **9**
4.1 Uniform consistency estimates 9
4.2 Comparison principle and discrete solvability 12
4.3 Uniform maximum-norm stability 12

5 Matrix structure and solver choice **15**
5.1 Weighted symmetry and the energy identity 15
5.2 Discrete coercivity and energy-error bounds 16
5.3 Solver implications of the proved matrix structure 17

6 Verification **17**
6.1 Evidence plan 17
6.2 Equation-level checks 17
6.3 Independent finite-element oracle 19
6.4 Separation of algebraic and discretisation error 20
6.5 Reproducibility and test suite 22

7 Numerical experiments **23**
7.1 Protocol, norms and order fitting 23
7.2 Global convergence of the two closures 23
7.3 Boundary error, interior error and finite-grid deviations 25
7.4 Matrix structure and weighted-energy norm 27
7.5 Measured cost at fixed accuracy thresholds 29
7.6 Extension: the pure-Neumann problem 31

8 Conclusions and limitations **32**

A Supplementary numerical data **34**
A.1 Nodal errors on every refinement level 34
A.2 Weighted energy errors and identity checks 34
A.3 Raw-matrix structure and algebraic residuals 34

B Pre-computation predictions and reconciliation **34**

C AI Transparency Log **35**

C.1 Supplementary five-gate audit record 35

D Individual Contribution Statement **35**

Abstract

We study how the right-edge closure affects finite difference solutions of a Poisson problem with Dirichlet data on three sides and Neumann or Robin data on the fourth. A backward derivative row is compared with a centred ghost-point condition eliminated into a boundary PDE row. A discrete comparison principle and a boundary-adapted barrier give mesh-independent maximum-norm estimates. Although both raw boundary defects are $O(h)$, their stability weights differ, yielding $O(h)$ and $O(h^2)$ global error bounds. A separate comparison proves a matching first-order lower bound for the exponential benchmark. Positive diagonal weights produce symmetric positive definite systems and discrete energy estimates. Six-mesh experiments confirm approximately first- and second-order convergence in the trapezoidal discrete L^2, nodal maximum and closure-dependent energy norms. Independent boundary-row, manufactured-solution and uneliminated-system checks test the assembly; tolerance sweeps and weighted CG cross-check the linear solves. Measured assembly-plus-solve costs are compared at fixed nodal-error thresholds. Independent FEM and pure-Neumann depth checks extend the verification. The conclusions distinguish proved bounds, observed rates and implementation-dependent timing.

Keywords: Poisson equation; mixed boundary conditions; finite differences; ghost points; comparison principle; error estimates; reproducibility.

1 Introduction

Steady heat conduction and electrostatic models combine prescribed values, prescribed outward derivatives and exchange conditions on different boundary segments. A numerical method must translate all of these conditions into equations consistent with its interior operator. The present report studies this issue through Direction 2 of the supplied Problem Pack: a reproducible Poisson problem on the unit square with Dirichlet values on three sides and Neumann or Robin data on the open right edge.

The direction prescribes Cartesian finite differences and two boundary closures. The first replaces the boundary PDE row by a backward derivative equation. The second eliminates a conceptual ghost point from a centred derivative equation and retains a PDE row at the boundary. Our question is how this change of boundary formulation affects the solution error, the matrix structure and the cost of achieving a named accuracy target. The two formulations differ in the equations they enforce and in their row scaling, as well as in derivative accuracy.

LeVeque [1] develops the finite difference framework for boundary value problems. Within that framework, consistency must be combined with stability before it implies convergence. For the present operators, both raw boundary rows have an $O(h)$ defect. We derive a maximum-norm stability estimate that weights the ghost boundary defect by an additional factor h, explaining its $O(h^2)$ global upper bound. The analysis uses the actual mixed boundary and its corner convention.

The exact function $u^* = e^{x+y}$ determines the forcing and boundary data independently of the assembled matrix. This permits exact error evaluation and a manufactured-solution verification study in the sense of Salari and Knupp [2]. The benchmark verifies numerical methods; it does not calibrate a model against an experimental device.

The analytical contributions are an explicit comparison principle, inverse positivity, uniform maximum-norm estimates, a sharp first-order maximum-error result for the exponential FO benchmark, and a weighted energy identity with discrete L^2 and energy bounds. The weighting is a standard diagonal symmetrisation; no new general CG algorithm is claimed. The computational study tests these results in all four boundary/closure configurations, separates algebraic and discretisation errors on a representative grid, and reports measured cost at fixed accuracy thresholds.

Section 2 states the model and regularity assumptions. Section 3 derives the discrete systems and the algorithm that assembles them. Section 4 establishes stability and convergence, and Section 5 the weighted matrix structure and the resulting solver choice. Section 6 collects the evidence that the implementation and the linear solves can be trusted. Section 7 presents the numerical experiments and the cost comparison, and Section 8 relates the measured results to the proved statements and states their limitations.

2 Mathematical model

2.1 Domain, boundary data and regularity

We now specify the selected mixed-boundary benchmark. All variables are nondimensional. Set

$$\begin{aligned} &\Omega = (0,1)^2, \\ &\Gamma_D = (\{0\} \times [0,1]) \cup ([0,1] \times \{0,1\}), \quad \Gamma_B = \{1\} \times (0,1). \end{aligned} \tag{2.1}$$

The governing equations are

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \Gamma_D, \\ \partial_n u + \beta u = s_\beta & \text{on } \Gamma_B. \end{cases} \tag{2.2}$$

The outward normal on Γ_B is $(1,0)$, so $\partial_n u = u_x$ there. The required variants are $(\beta, s_\beta) = (0, q)$ and $(2, r)$. Every corner belongs to the Dirichlet node set. In the weak formulation, traces are understood almost everywhere, and isolated corner points have zero boundary measure.

Physical interpretation of the boundary data. For a homogeneous conductor with constant thermal conductivity $k > 0$, write the dimensional position and temperature as $\boldsymbol{X} = L(x, y)$ and $T(\boldsymbol{X}) = T_{\text{ref}} + \Delta T\, u(x, y)$, where $L > 0$ and $\Delta T > 0$. Fourier's law gives the outward heat flux

$$j_n = -k\, \partial_{n_{\boldsymbol{X}}} T = -\frac{k \Delta T}{L}\, \partial_n u.$$

Thus the Neumann datum q prescribes a scaled normal derivative: $j_n = -(k\Delta T/L)q$; $q = 0$ describes an insulated edge. For convection to a surrounding fluid at temperature T_∞, with heat-transfer coefficient $H > 0$, the boundary balance is $-k\, \partial_{n_{\boldsymbol{X}}} T = H(T - T_\infty)$. Setting $u_\infty = (T_\infty - T_{\text{ref}})/\Delta T$ yields

$$\partial_n u + \mathrm{Bi}\, u = \mathrm{Bi}\, u_\infty, \qquad \mathrm{Bi} = \frac{HL}{k}. \tag{2.3}$$

This is the standard Biot-number interpretation of a convective Robin condition [**?**]. Under this particular scaling, $\beta = 2$ corresponds to $\mathrm{Bi} = 2$, and the Robin load is $r = 2u_\infty$; r itself is not a heat flux. The manufactured data specify a numerical benchmark, without identifying a particular material or device.

Figure 1 fixes the boundary ownership used below.

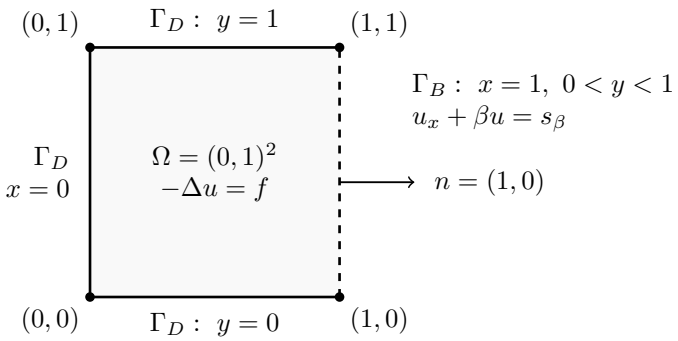


Figure 1: Boundary ownership for the selected benchmark. The solid edges and all four filled corner points carry Dirichlet data. The dashed open right edge carries the derivative/Robin condition; its endpoints are excluded from the boundary-row unknowns. This analytical schematic fixes the outward-normal sign and the corner convention before any computation.

Assumption 2.1 (Smooth solution and consistent data)**.** The coefficient $\beta \geq 0$ is constant and independent of the mesh. The exact solution u extends to a C^6 function on an open neighbourhood of $\overline{\Omega}$. The data satisfy $f = -\Delta u$ on $\overline{\Omega}$, $g = u|_{\Gamma_D}$ and $s_\beta = (u_x + \beta u)|_{\Gamma_B}$. Dirichlet values used in the analysis are imposed exactly. For $\ell = 2, 3, 4$, write

$$M_\ell = \max\{\|\partial_x^\ell u\|_{L^\infty(\overline{\Omega})}, \|\partial_y^\ell u\|_{L^\infty(\overline{\Omega})}\}. \tag{2.4}$$

All asymptotic remainders below are uniform over the relevant grid nodes; their constants may depend on derivative bounds for u, but not on N, h, i or j.

This assumption is verified for the manufactured benchmark. It is an assumption about the solution, rather than a regularity conclusion for arbitrary mixed data; general mixed-boundary problems may have corner singularities.

For a C^1 solution with smooth edge data, a necessary endpoint compatibility is

$$\begin{aligned} \lim_{y \downarrow 0} s_\beta(y) &= \partial_x g(1,0) + \beta g(1,0), \\ \lim_{y \uparrow 1} s_\beta(y) &= \partial_x g(1,1) + \beta g(1,1). \end{aligned} \tag{2.5}$$

Here $\partial_x g$ is the tangential derivative on the bottom or top Dirichlet edge. The identities follow by taking limits of the right-edge condition and differentiating the Dirichlet trace along its horizontal edge. They are necessary conditions for a smooth solution, not extra discrete corner equations. The exponential data below satisfy both.

2.2 Manufactured benchmark

For the prescribed exact solution $u^*(x, y) = e^{x+y}$, direct differentiation gives

$$\begin{aligned} f(x,y) &= -2e^{x+y}, \\ g(0,y) &= e^y, \qquad g(x,0) = e^x, \qquad g(x,1) = e^{x+1}, \\ q(y) &= e^{1+y}, \qquad r(y) = 3e^{1+y}, \quad s_\beta(y) = (1+\beta)e^{1+y}. \end{aligned} \tag{2.6}$$

The corner values at $(0,0), (0,1), (1,0), (1,1)$ are $1, e, e, e^2$, respectively, and $M_2 = M_3 = M_4 = e^2$. The forcing is derived from the continuous solution independently of the discrete matrix. Consequently, substituting sampled exact values into the discrete equations retains the defects that the verification study must detect. Manufactured solutions support code verification through exact error evaluation and systematic refinement [2]; this benchmark does not provide experimental validation of a heat-transfer model.

2.3 Weak formulation and uniqueness

Let Tr denote the Sobolev trace and set

$$V_0 = \{v \in H^1(\Omega) : \operatorname{Tr} v = 0 \text{ on } \Gamma_D\}, \qquad V_g = \{v \in H^1(\Omega) : \operatorname{Tr} v = g \text{ on } \Gamma_D\}.$$

For $u \in V_g$, the weak formulation is

$$a_\beta(u, v) = \ell(v) \qquad (v \in V_0), \tag{2.7}$$

where

$$a_\beta(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx\, dy + \beta \int_{\Gamma_B} uv \, ds, \tag{2.8}$$
$$\ell(v) = \int_\Omega f v \, dx\, dy + \int_{\Gamma_B} s_\beta v \, ds.$$

The boundary values in these integrals denote traces. This sign convention agrees with the standard weak treatment of Neumann and Robin data in [3].

Lemma 2.2 (Coercivity induced by the left Dirichlet edge)**.** *For every $v \in V_0$,*

$$\|v\|^2_{L^2(\Omega)} \le \tfrac{1}{2}\|v_x\|^2_{L^2(\Omega)}, \qquad \|\operatorname{Tr} v\|^2_{L^2(\Gamma_B)} \le \|v_x\|^2_{L^2(\Omega)}. \tag{2.9}$$

Consequently, for $\beta \ge 0$,

$$a_\beta(v, v) \ge \tfrac{2}{3}\|v\|^2_{H^1(\Omega)}, \qquad \|v\|^2_{H^1(\Omega)} := \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)}. \tag{2.10}$$

Proof. For almost every y, the one-dimensional H^1 slice has an absolutely continuous representative and its left trace is zero. Hence $v(x, y) = \int_0^x v_x(t, y)\, dt$. Cauchy–Schwarz gives

$$|v(x, y)|^2 \le x \int_0^x |v_x(t, y)|^2 \, dt.$$

Integrate first in x, interchange the nonnegative integrals, and then integrate in y:

$$\begin{aligned}\|v\|^2_{L^2(\Omega)} &\le \int_0^1 \int_0^1 \left(\int_t^1 x \, dx\right) |v_x(t, y)|^2 \, dt\, dy \\ &= \int_0^1 \int_0^1 \tfrac{1-t^2}{2} |v_x(t, y)|^2 \, dt\, dy \le \tfrac{1}{2}\|v_x\|^2_{L^2(\Omega)}.\end{aligned}$$

At $x = 1$, the same representation yields $|v(1, y)|^2 \le \int_0^1 |v_x(t, y)|^2 \, dt$, proving the trace bound. Finally $\|v\|^2_{H^1} \le \frac{3}{2}\|\nabla v\|^2_{L^2}$, while $a_\beta(v, v) \ge \|\nabla v\|^2_{L^2}$ because $\beta \ge 0$. Combining these inequalities gives (2.10). □

Proposition 2.3 (Solvability for the stated smooth data)**.** *Under Assumption 2.1, the function u is the unique weak solution of (2.7). In particular, both required benchmark variants have the unique solution $u^* = e^{x+y}$.*

Proof. The assumed trace of u shows $u \in V_g$. Green's identity, applicable to this smooth u and $v \in H^1(\Omega)$, gives

$$\int_\Omega \nabla u \cdot \nabla v = \int_\Omega f v + \int_{\partial\Omega} \partial_n u \, v = \int_\Omega f v + \int_{\Gamma_B} (s_\beta - \beta u) v \qquad (v \in V_0).$$

The contribution from Γ_D vanishes because the trace of v is zero there. Moving the Robin term to the left proves (2.7) and establishes existence.

For uniqueness, let $u_1, u_2 \in V_g$ solve (2.7) and put $w = u_1 - u_2 \in V_0$. Subtraction gives $a_\beta(w, v) = 0$ for all $v \in V_0$. Taking $v = w$ yields

$$\|\nabla w\|^2_{L^2(\Omega)} + \beta\|\operatorname{Tr} w\|^2_{L^2(\Gamma_B)} = 0.$$

Lemma 2.2 gives $\frac{2}{3}\|w\|^2_{H^1(\Omega)} \le a_\beta(w, w) = 0$. Therefore $w = 0$ in $H^1(\Omega)$, and $u_1 = u_2$. □

The Dirichlet boundary removes the constant nullspace even when $\beta = 0$. A pure-Neumann compatibility condition and a pressure gauge are therefore not needed for this problem. No existence claim for arbitrary nonsmooth data is used in the subsequent analysis.

3 Finite difference discretisation

3.1 Grid, unknowns and norms

For $N \geq 2$, let $h = 1/N$, $x_i = ih$ and $y_j = jh$. Define the full index set $\mathcal{X}_h = \{0, \dots, N\}^2$ and the disjoint node sets

$$\begin{aligned} \mathcal{I}_h &= \{(i,j) : 1 \leq i, j \leq N-1\}, \\ \mathcal{B}_h &= \{(N,j) : 1 \leq j \leq N-1\}, \\ \mathcal{U}_h &= \mathcal{I}_h \cup \mathcal{B}_h, \qquad\qquad \mathcal{D}_h = \mathcal{X}_h \setminus \mathcal{U}_h. \end{aligned} \tag{3.1}$$

A grid function is a real array on $\mathcal{X}_h$. A vector on $\mathcal{U}_h$ will be extended by zero on $\mathcal{D}_h$ whenever used in an error estimate or quadratic form. The ordering is

$$k(i,j) = (i-1) + N(j-1), \qquad 0 \leq k < N(N-1) = n_{\mathrm{dof}}. \tag{3.2}$$

There are $(N-1)^2$ interior and $N-1$ right-edge unknowns. Ghost values are not degrees of freedom.

For a full grid function V, write

$$\begin{aligned} \|V\|_{\infty,h} &= \max_{(i,j)\in\mathcal{X}_h} |V_{i,j}|, \\ \|V\|_{2,h}^2 &= h^2 \sum_{i=0}^{N}\sum_{j=0}^{N} \omega_i \omega_j V_{i,j}^2, \qquad \omega_0 = \omega_N = \tfrac{1}{2}, \quad \omega_i = 1 \ (1 \leq i \leq N-1). \end{aligned} \tag{3.3}$$

For arrays on the interior and open right edge, respectively, also define

$$\|F\|_{I,h}^2 = h^2 \sum_{(i,j)\in\mathcal{I}_h} F_{i,j}^2, \quad \|S\|_{\Gamma,h}^2 = h \sum_{j=1}^{N-1} S_j^2. \tag{3.4}$$

Subscripts ∞, I and ∞, B will denote maxima over $\mathcal{I}_h$ and $\mathcal{B}_h$.

3.2 Interior stencil and first-order closure

The centred second differences are

$$(D_{xx}^h V)_{i,j} = \frac{V_{i+1,j} - 2V_{i,j} + V_{i-1,j}}{h^2}, \qquad (D_{yy}^h V)_{i,j} = \frac{V_{i,j+1} - 2V_{i,j} + V_{i,j-1}}{h^2}.$$

The usual Taylor derivation of these formulas is recalled and quantified in Lemma 4.1; see also [4]. The interior operator is $L_h = -D_{xx}^h - D_{yy}^h$, giving

$$(L_h U)_{i,j} = \frac{4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1}}{h^2} = f_{i,j}, \qquad (i,j) \in \mathcal{I}_h. \tag{3.5}$$

Its neighbour at $i = N$ remains an unknown.

The first-order (FO) closure replaces the boundary PDE row with

$$(B_h^\beta U)_j = \left(\frac{1}{h} + \beta\right) U_{N,j} - \frac{1}{h} U_{N-1,j} = s_{\beta,j}, \qquad 1 \leq j \leq N-1. \tag{3.6}$$

This equation has no tangential coupling. Its diagonal is $1/h$ for Neumann data and $1/h + 2$ for the prescribed Robin data.

3.3 Ghost-point closure

Define the eliminated ghost (G) operator by

$$(G_h^\beta V)_j = \frac{4V_{N,j} - 2V_{N-1,j} - V_{N,j-1} - V_{N,j+1}}{h^2} + \frac{2\beta}{h} V_{N,j}. \tag{3.7}$$

Proposition 3.1 (Exact ghost elimination)**.** *For a fixed boundary index j, the centred boundary condition*

$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} + \beta U_{N,j} = s_{\beta,j} \tag{3.8}$$

and the boundary five-point PDE row

$$4U_{N,j} - U_{N-1,j} - U_{N+1,j} - U_{N,j-1} - U_{N,j+1} = h^2 f_{N,j} \tag{3.9}$$

are equivalent to the single retained equation

$$(G_h^\beta U)_j = f_{N,j} + \frac{2s_{\beta,j}}{h} \tag{3.10}$$

together with the uniquely determined auxiliary value

$$U_{N+1,j} = U_{N-1,j} + 2hs_{\beta,j} - 2h\beta U_{N,j}. \tag{3.11}$$

Proof. Since $h > 0$, multiplying (3.8) by $2h$ and solving for its ghost value gives (3.11). Substitution into (3.9) gives

$$(4 + 2h\beta)U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1} = h^2 f_{N,j} + 2hs_{\beta,j}.$$

Dividing by h^2 proves (3.10). Conversely, given retained values satisfying (3.10), define the ghost value by (3.11). The centred boundary condition then holds identically, and multiplication of (3.10) by h^2 and reverse substitution recovers (3.9). This proves both directions and uniqueness of the auxiliary value. □

Thus the ghost method imposes a boundary PDE row after elimination, whereas the first method imposes a derivative row. Both have exactly $N(N-1)$ retained equations and unknowns.

3.4 Dirichlet elimination and matrix scaling

For $c \in \{\mathrm{FO}, \mathrm{G}\}$, let A_h^c denote the reduced matrix obtained from (3.5) and the corresponding raw boundary row (3.6) or (3.10). If $\mathcal{A}_h^c$ is the stencil matrix with columns for all nodes, the reduced system is

$$A_h^c \boldsymbol{U}_h^c = \boldsymbol{b}_h^c, \qquad A_h^c = (\mathcal{A}_h^c)_{\mathcal{U}_h \mathcal{U}_h}, \qquad \boldsymbol{b}_h^c = \boldsymbol{F}_h^c - (\mathcal{A}_h^c)_{\mathcal{U}_h \mathcal{D}_h} \boldsymbol{g}_h. \tag{3.12}$$

Here $\boldsymbol{F}_h^c$ is f in the interior and is s_β (FO) or $f + 2s_\beta/h$ (G) on the right edge. In particular, a known entry $a_{kD} g_D$ contributes $-a_{kD} g_D$ to the right-hand side. The ghost rows at $j = 1$ and $j = N - 1$ receive $g(1,0)/h^2$ and $g(1,1)/h^2$, respectively. For $N = 2$ both contributions enter the same row. Neither corner is assigned a derivative row.

Across the interface between $\mathcal{I}_h$ and $\mathcal{B}_h$, the interior coefficient is $-1/h^2$, while the reverse coefficient is $-1/h$ (FO) or $-2/h^2$ (G). Hence both raw matrices are nonsymmetric for $N \geq 2$. Their symmetrised forms will be established in Proposition 5.1, rather than inferred from the continuous Laplacian.

Proposition 3.2 (Analytical sparsity counts)**.** *For $N \geq 2$ and $\beta \geq 0$, the numbers of nonzero entries in the reduced matrices are*

$$\begin{aligned} \operatorname{nnz}(A_h^{\mathrm{FO}}) &= (N-1)(5N-6), \\ \operatorname{nnz}(A_h^{\mathrm{G}}) &= (N-1)(5N-6) + 2(N-2). \end{aligned} \tag{3.13}$$

Both counts grow linearly with $n_{\mathrm{dof}} = N(N-1)$.

Proof. Every interior row contains a diagonal and an east unknown, contributing $2(N-1)^2$ entries. West unknowns occur for $i \geq 2$, contributing $(N-2)(N-1)$ entries. Each of the two vertical directions contributes $(N-2)(N-1)$ entries. Thus the interior rows contain $2(N-1)^2 + 3(N-2)(N-1) = (N-1)(5N-8)$ entries. Every boundary row contains its diagonal and west unknown, adding $2(N-1)$ entries. The FO closure has no further entries, giving its count in (3.13). The G closure also has $N-2$ unknown neighbours in each vertical direction, giving the additional $2(N-2)$ entries. All counted off-diagonals are strictly negative and all diagonals are strictly positive; none vanish for the stated β. Dirichlet entries have already been removed. These counts include $N = 2$, when the additional vertical-boundary count is zero. Every unknown contributes a nonzero diagonal, giving $\operatorname{nnz}(A_h^c) \geq n_{\mathrm{dof}}$. The formulas also give $\operatorname{nnz}(A_h^c) \leq 5n_{\mathrm{dof}}$ for both closures and all $N \geq 2$. These two bounds prove the stated linear growth. □

For both $\beta = 0$ and $\beta = 2$, the assembled matrices in the four-grid refinement data have exactly the nonzero counts in (3.13): the FO/G pairs are $(1110, 1138)$ at $N = 16$, $(4774, 4834)$ at $N = 32$, $(19782, 19906)$ at $N = 64$, and $(80518, 80770)$ at $N = 128$. This agreement checks that the implemented connectivity matches the analytical stencil counts. A nonzero-count check alone does not verify the coefficient values or the right-hand side. The counts refer to the reduced matrices before factorisation.

3.5 Algorithm and implementation

The project-specific code implements grid ownership, the ordering (3.2), both boundary closures, exact Dirichlet elimination, error norms, componentwise residuals, row weights and the edge-sum energy. Sparse entries are assembled using array masks and vectorised coordinate construction in COO and converted to CSR; the direct solve converts the matrix to CSC. Assembly, reconstruction, exact-vector evaluation and row-weight construction contain no Python loops over grid points. The discretisation, boundary closures, Dirichlet elimination, error diagnostics and weighted structure are implemented by the authors, because they are the object of the verification. NumPy supplies array operations and a dense small-system solve; SciPy supplies sparse storage, SuperLU, GMRES, ILU, CG and the symmetric eigenvalue routine; Matplotlib supplies figures. These interfaces are delegated numerical services rather than separately implemented solvers: the aim is to test the discretisation, not to reimplement sparse factorisation or Krylov methods.

The verification of this algorithm is the subject of Section 6; the experiments that use it are reported in Section 7.

4 Stability and error analysis

4.1 Uniform consistency estimates

Let u_h denote u sampled on the full grid and let $\boldsymbol{u}_h$ be its restriction to the unknowns. Define

$$\boldsymbol{\tau}_h^c = A_h^c \boldsymbol{u}_h - \boldsymbol{b}_h^c. \tag{4.1}$$

1. **Input:** $N \geq 2$, closure $c \in \{\mathrm{FO}, \mathrm{G}\}$, $\beta \in \{0, 2\}$, and independent functions u^*, f, s_β. Set $h = 1/N$ and enumerate $\mathcal{U}_h$ by (3.2).
2. For each interior node, assemble (3.5). For each open right-edge node, assemble (3.6) or (3.10). Do not assign a derivative row to a corner.
3. For each known Dirichlet neighbour with coefficient a_{kD}, add $-a_{kD}g_D$ to the right-hand side. Store only unknown columns.
4. Convert COO entries to CSR, sum duplicate entries, and check the dimensions and finiteness of $A_h^c, \boldsymbol{b}_h^c$.
5. Solve $A_h^c \widetilde{\boldsymbol{U}}_h^c = \boldsymbol{b}_h^c$ using the fixed sparse direct backend. Reconstruct all grid values, retaining exact Dirichlet values.
6. Evaluate the global norms (3.3), restricted boundary/interior errors, residuals (6.1), matrix asymmetry and, where required, the weighted energy error.
7. **Output:** solution, diagnostics and measured times. The manufactured error is evaluated against u^*, not inferred from the residual.

Algorithm 3.1: A mixed-boundary finite difference solve and its diagnostics

Exact Dirichlet elimination means that these entries are precisely

$$\begin{aligned} \tau_{I,i,j} &= (L_h u_h)_{i,j} - f_{i,j}, \\ \tau_{B,j}^{\mathrm{FO}} &= (B_h^\beta u_h)_j - s_{\beta,j}, \\ \tau_{B,j}^{\mathrm{G}} &= (G_h^\beta u_h)_j - f_{N,j} - 2s_{\beta,j}/h. \end{aligned} \tag{4.2}$$

This definition fixes the row scaling in every subsequent estimate.

Lemma 4.1 (Local defects and uniform bounds)**.** *Under Assumption 2.1, the following expansions hold, with derivatives evaluated at the relevant node:*

$$\tau_I = -\frac{h^2}{12}(u_{xxxx} + u_{yyyy}) + O(h^4), \tag{4.3}$$

$$\tau_B^{\mathrm{FO}} = -\frac{h}{2}u_{xx} + \frac{h^2}{6}u_{xxx} + O(h^3), \tag{4.4}$$

$$\tau_B^{\mathrm{G}} = \frac{h}{3}u_{xxx} - \frac{h^2}{12}(u_{xxxx} + u_{yyyy}) + O(h^3). \tag{4.5}$$

For every $N \geq 2$, the explicit bounds are

$$\begin{aligned} \|\tau_I\|_{\infty,I} &\leq \frac{M_4}{6}h^2, \\ \|\tau_B^{\mathrm{FO}}\|_{\infty,B} &\leq \frac{M_2}{2}h, \\ \|\tau_B^{\mathrm{G}}\|_{\infty,B} &\leq \frac{M_3}{3}h + \frac{M_4}{12}h^2. \end{aligned} \tag{4.6}$$

Proof. Taylor's theorem along the x grid line gives, at an interior node,

$$u(x_i \pm h, y_j) = \sum_{\ell=0}^{5} \frac{(\pm h)^\ell}{\ell!}\partial_x^\ell u(x_i, y_j) + R_\pm, \qquad |R_\pm| \leq C_6 h^6/720,$$

where C_6 is a uniform bound for the pure sixth derivatives on $\overline{\Omega}$. Adding the expansions cancels all odd terms and yields $D_{xx}^h u = u_{xx} + h^2 u_{xxxx}/12 + O(h^4)$. The same computation in y, followed

by $f = -\Delta u$, proves (4.3). Using Taylor's theorem only through order three instead bounds each centred second-derivative defect by $M_4 h^2/12$, proving the first inequality in (4.6).

At a right-edge node, write $u = u(1, y_j)$ and $u_W = u(1-h, y_j)$. Taylor's theorem gives

$$u_W = u - hu_x + \frac{h^2}{2}u_{xx} - \frac{h^3}{6}u_{xxx} + O(h^4).$$

Substitution in $(u - u_W)/h + \beta u - (u_x + \beta u)$ proves (4.4). More directly, the second-order Taylor remainder has magnitude at most $M_2 h^2/2$, giving $|\tau_B^{\mathrm{FO}}| \le M_2 h/2$.

For the ghost row, expand the inward and tangential terms separately:

$$\begin{aligned} 2u - 2u_W &= 2hu_x - h^2 u_{xx} + \frac{h^3}{3}u_{xxx} - \frac{h^4}{12}u_{xxxx} + O(h^5), \\ 2u - u(1, y_j - h) - u(1, y_j + h) &= -h^2 u_{yy} - \frac{h^4}{12}u_{yyyy} + O(h^6). \end{aligned}$$

After division by h^2 and addition of $2\beta u/h$, this becomes

$$G_h^\beta u_h = \frac{2}{h}(u_x + \beta u) - \Delta u + \frac{h}{3}u_{xxx} - \frac{h^2}{12}(u_{xxxx} + u_{yyyy}) + O(h^3).$$

The first two terms equal $2s_\beta/h + f$, proving (4.5). For its uniform bound, use the exact decomposition

$$\tau_B^{\mathrm{G}} = \left[\frac{2}{h^2}(u - u_W - hu_x) + u_{xx}\right] + (u_{yy} - D_{yy}^h u_h).$$

The third-order Taylor remainder for u_W bounds the bracket by $M_3 h/3$. The centred tangential defect is at most $M_4 h^2/12$. The triangle inequality proves the final bound. Every Taylor segment used here lies in $\overline{\Omega}$, including when $N = 2$; bounded derivatives therefore make all constants independent of the node and mesh. □

Corollary 4.2 (Exact benchmark identities)**.** *For $u^* = e^{x+y}$, the defects satisfy*

$$\begin{aligned} \frac{\tau_I}{u^*} &= 2 + \frac{4 - 2e^h - 2e^{-h}}{h^2}, \\ \frac{\tau_B^{\mathrm{FO}}}{u^*} &= \frac{1 - e^{-h}}{h} - 1, \\ \frac{\tau_B^{\mathrm{G}}}{u^*} &= \frac{4 - 3e^{-h} - e^h}{h^2} - \frac{2}{h} + 2. \end{aligned} \tag{4.7}$$

The limits of these ratios divided by h^2, h and h, respectively, are $-1/6$, $-1/2$ and $1/3$ as $h \to 0$.

Proof. Relative to a node's exact value, its east and north neighbours contribute e^h, while its west and south neighbours contribute e^{-h}. Insert these ratios and the data (2.6) into (4.2). The Robin terms cancel exactly. Expanding $e^{\pm h}$ through the first nonzero defect term gives the stated limits. □

Although the centred derivative in (3.8) has an $O(h^2)$ defect, the raw eliminated PDE row has an $O(h)$ defect. The global effect of that row must be established using its boundary stability weight.

4.2 Comparison principle and discrete solvability

Lemma 4.3 (Discrete comparison principle)**.** *Let $N \geq 2$ and $\beta \geq 0$. Suppose a grid function V satisfies $V \geq 0$ on $\mathcal{D}_h$ and $L_h V \geq 0$ on $\mathcal{I}_h$. If either $B_h^\beta V \geq 0$ on $\mathcal{B}_h$ or $G_h^\beta V \geq 0$ there, then $V \geq 0$ on the whole grid.*

Proof. Suppose instead that the minimum of V is $m < 0$. The finite grid contains a node attaining m, which cannot belong to $\mathcal{D}_h$. At an interior minimum $P = (i, j)$,

$$0 \leq (L_h V)_P = \frac{1}{h^2} \sum_{Q \in \{(i-1,j),(i+1,j),(i,j-1),(i,j+1)\}} (m - V_Q) \leq 0.$$

Each summand is nonpositive and their sum is zero, so every neighbour equals m. In particular, the west neighbour is another minimum. Repeat this argument with decreasing i until reaching $(0, j)$, contradicting the nonnegative Dirichlet value.

If the initial minimum is at (N, j), the FO row gives

$$0 \leq (B_h^\beta V)_j = \frac{m - V_{N-1,j}}{h} + \beta m \leq 0.$$

For $\beta > 0$ the final inequality is strict, an immediate contradiction. For $\beta = 0$, equality forces $V_{N-1,j} = m$; since $N \geq 2$, this is an interior node and the preceding argument applies. For the ghost row,

$$0 \leq (G_h^\beta V)_j = \frac{2(m - V_{N-1,j}) + (m - V_{N,j-1}) + (m - V_{N,j+1})}{h^2} + \frac{2\beta}{h} m \leq 0.$$

Again $\beta > 0$ gives a contradiction. If $\beta = 0$, every weighted nonpositive term must vanish; in particular $V_{N-1,j} = m$, so the same interior argument applies. These cases cover every possible minimum. □

We use the standard inverse-positive characterisation of nonsingular M-matrices; see Berman and Plemmons [5]. The properties needed for these particular matrices are proved directly below.

Proposition 4.4 (Invertibility and inverse positivity)**.** *For $N \geq 2$, $\beta \geq 0$ and either closure, A_h^c is invertible, has nonpositive off-diagonal entries, and satisfies $(A_h^c)^{-1} \geq 0$ entrywise. Thus it has the inverse-positive characterisation of a nonsingular M-matrix.*

Proof. The sign of every off-diagonal entry follows directly from the stencils. If $A_h^c \boldsymbol{V} = 0$, extend $\boldsymbol{V}$ by zero on $\mathcal{D}_h$. Its full-grid interior and boundary rows are zero. Lemma 4.3 applied to both V and $-V$ gives $V = 0$. The square matrix A_h^c therefore has a trivial kernel and is invertible. For any vector $\boldsymbol{d} \geq 0$, its solution $\boldsymbol{V} = (A_h^c)^{-1}\boldsymbol{d}$, extended by zero, satisfies the hypotheses of that lemma and hence is nonnegative. Taking $\boldsymbol{d}$ to be each coordinate vector proves that every column of $(A_h^c)^{-1}$ is nonnegative. □

4.3 Uniform maximum-norm stability

Lemma 4.5 (A boundary-adapted barrier)**.** *Let $\psi_{i,j} = \psi(x_i)$, where $\psi(x) = 1 + x - x^2/4$. For $N \geq 2$, $1 \leq \psi_{i,j} \leq 7/4$, and the full-grid operators satisfy*

$$\begin{aligned} L_h \psi &= \tfrac{1}{2}, \\ B_h^\beta \psi &= \tfrac{1}{2} + \tfrac{h}{4} + \tfrac{7\beta}{4} \geq \tfrac{1}{2}, \\ G_h^\beta \psi &= \tfrac{1}{h} + \tfrac{1}{2} + \tfrac{7\beta}{2h} \geq \tfrac{1}{h}. \end{aligned} \tag{4.8}$$

Proof. Since $\psi'(x) = 1 - x/2 > 0$ on $[0,1]$, its endpoint values give the stated range. The centred differences are exact on this quadratic and ψ does not depend on y, so $L_h\psi = -\psi'' = 1/2$. Moreover $\psi(1) = 7/4$ and $\psi(1) - \psi(1-h) = h/2 + h^2/4$. Substitution into the FO row yields the second identity. In the ghost row the tangential differences vanish, leaving $2[\psi(1) - \psi(1-h)]/h^2 + 2\beta\psi(1)/h$, which gives the third identity. The inequalities use $\beta \geq 0$. □

Proposition 4.6 (Uniform stability in the raw row scaling)**.** *Let $N \geq 2$, $\beta \geq 0$ and $V = 0$ on $\mathcal{D}_h$. Suppose $L_h V = F$ on $\mathcal{I}_h$ and the selected boundary operator applied to V equals S on $\mathcal{B}_h$. Set $F_0 = \|F\|_{\infty,I}$ and $S_0 = \|S\|_{\infty,B}$. Then*

$$\|V\|_{\infty,h} \leq \tfrac{7}{2}(F_0 + S_0) \qquad \text{for FO,} \tag{4.9}$$

$$\|V\|_{\infty,h} \leq \tfrac{7}{4}(2F_0 + hS_0) \qquad \text{for G.} \tag{4.10}$$

The constants are independent of N and $\beta \geq 0$.

Proof. Use the full-grid barrier of Lemma 4.5. For FO, set $a = 2(F_0 + S_0)$. On $\mathcal{D}_h$, $a\psi \pm V = a\psi \geq 0$. In the interior, $L_h(a\psi \pm V) = a/2 \pm F \geq a/2 - F_0 \geq 0$. On the right edge, $B_h^\beta(a\psi \pm V) \geq a/2 - S_0 \geq 0$. Lemma 4.3 gives $a\psi - V \geq 0$ and $a\psi + V \geq 0$, so $|V| \leq a\psi \leq 7a/4$, proving (4.9).

For G, choose $a = 2F_0 + hS_0$. The Dirichlet values are again nonnegative, the interior rows satisfy $a/2 - F_0 = hS_0/2 \geq 0$, and the boundary rows satisfy $G_h^\beta(a\psi \pm V) \geq a/h - S_0 = 2F_0/h \geq 0$. The same comparison argument yields $|V| \leq 7a/4$ and proves (4.10). The barrier has its own nonzero Dirichlet values; the proof uses full-grid operators, not a reduced matrix multiplication that discards those values. □

Theorem 4.7 (Maximum-norm convergence)**.** *Under Assumption 2.1, let U_h^c be the unique discrete solution, including its exact Dirichlet values, and set $e_h^c = U_h^c - u_h$. For all $N \geq 2$,*

$$\|e_h^{\mathrm{FO}}\|_{\infty,h} \leq \tfrac{7}{2}\left(\tfrac{M_4}{6}h^2 + \tfrac{M_2}{2}h\right), \tag{4.11}$$

$$\|e_h^{\mathrm{G}}\|_{\infty,h} \leq \tfrac{7}{4}\left(\tfrac{M_4+M_3}{3}h^2 + \tfrac{M_4}{12}h^3\right). \tag{4.12}$$

Consequently, these maximum-norm errors are $O(h)$ and $O(h^2)$, respectively, as $N \to \infty$.

Proof. Proposition 4.4 gives existence and uniqueness of each discrete solution. On $\mathcal{D}_h$, $e_h^c = 0$. Subtraction of (4.1) from the discrete system gives

$$A_h^c \boldsymbol{e}_h^c = -\boldsymbol{\tau}_h^c. \tag{4.13}$$

Thus its full-grid interior equation is $L_h e_h^c = -\tau_I$ and its boundary equation is $B_h^\beta e_h^{\mathrm{FO}} = -\tau_B^{\mathrm{FO}}$ or $G_h^\beta e_h^{\mathrm{G}} = -\tau_B^{\mathrm{G}}$. Apply Proposition 4.6 and then Lemma 4.1 to obtain (4.11) and (4.12). Since $h \leq 1/2$ and M_ℓ is independent of h, the first bound is at most $\frac{7}{2}(M_4/12 + M_2/2)h$, and the second at most $\frac{7}{4}[(M_4 + M_3)/3 + M_4/24]h^2$. This proves uniform convergence with the stated upper orders. □

Theorem 4.8 (Sharp first-order maximum error for the exponential benchmark)**.** *Let $u^* = e^{x+y}$ and fix $\beta \geq 0$. For the exact FO discrete solution and every $N \geq 2$,*

$$\max_{1 \leq j \leq N-1} |e_{N,j}^{\mathrm{FO}}| \geq C_\beta h, \qquad C_\beta := \frac{e \sinh \pi}{2\sqrt{2}\, m_\beta} > 0, \quad m_\beta := \pi \cosh \pi + \beta \sinh \pi. \tag{4.14}$$

Together with Theorem 4.7, this implies $\|e_h^{\mathrm{FO}}\|_{\infty,h} = \Theta(h)$ and $\max_j |e_{N,j}^{\mathrm{FO}}| = \Theta(h)$ for each fixed β, in particular for both required variants.

Proof. For the exponential, the interior identity (4.7) and $\cosh h - 1 \geq h^2/2$ give

$$\frac{\tau_I}{u^*} = 2 - \frac{4(\cosh h - 1)}{h^2} \leq 0.$$

At a boundary node, cancellation of the Robin term and the fundamental theorem of calculus give the exact formula

$$-\tau_{B,j}^{\mathrm{FO}} = \frac{1}{h}\int_{1-h}^{1} (z - 1 + h) u_{xx}^*(z, y_j)\, dz.$$

Since $y_j \geq h$ and $z \geq 1 - h$, $u_{xx}^*(z, y_j) = e^{z+y_j} \geq e$. The integral of $z - 1 + h$ is $h^2/2$, so $-\tau_{B,j}^{\mathrm{FO}} \geq eh/2$.

Set $\phi_{i,j} = \sinh(\pi x_i)\sin(\pi y_j)$. It vanishes on $\mathcal{D}_h$ and is nonnegative on the full grid. Direct use of the addition formulas yields

$$(L_h\phi)_{i,j} = \frac{4 - 2\cosh(\pi h) - 2\cos(\pi h)}{h^2}\phi_{i,j} \leq 0.$$

The sign follows from $\cosh t + \cos t = 2\sum_{k=0}^{\infty} t^{4k}/(4k)! \geq 2$ for real t. On the right edge,

$$(B_h^{\beta}\phi)_j = \left[\frac{\sinh\pi - \sinh(\pi(1-h))}{h} + \beta\sinh\pi\right]\sin(\pi y_j) \leq m_\beta.$$

Indeed the backward quotient equals the average of $\pi\cosh(\pi x)$ over $[1 - h, 1]$, which is at most $\pi\cosh\pi$, and $0 \leq \sin(\pi y_j) \leq 1$.

Choose $a = e/(2m_\beta) > 0$ and define $V = e_h^{\mathrm{FO}} - ah\phi$. It has zero Dirichlet values. The error equation gives

$$L_h V = -\tau_I - ahL_h\phi \geq 0, \qquad B_h^{\beta} V = -\tau_B^{\mathrm{FO}} - ahB_h^{\beta}\phi \geq eh/2 - ahm_\beta = 0.$$

Lemma 4.3 implies $e_h^{\mathrm{FO}} \geq ah\phi$. Take a boundary index j closest to $N/2$. It belongs to $\{1, \ldots, N-1\}$ and satisfies $|y_j - 1/2| \leq h/2 \leq 1/4$, so $\sin(\pi y_j) \geq 1/\sqrt{2}$. Therefore $e_{N,j}^{\mathrm{FO}} \geq ah\sinh\pi/\sqrt{2} = C_\beta h$. The maximum-norm upper bound in Theorem 4.7 also bounds the boundary maximum by a constant times h. These two-sided bounds prove both $\Theta(h)$ statements. □

Remark 4.9 (Rates, bounds and measured errors). The factor h multiplying the ghost boundary defect in (4.10) converts its $O(h)$ local size into an $O(h^2)$ contribution to the solution error. Theorem 4.7 gives rigorous a priori upper bounds, but does not identify an asymptotic leading-error coefficient.

For the exponential benchmark, let $E_h^{\mathrm{FO}} = \|e_h^{\mathrm{FO}}\|_{\infty,h}$ and define

$$B_{\mathrm{FO}}(h) = \frac{7e^2}{2}\left(\frac{h^2}{6} + \frac{h}{2}\right).$$

Since $M_2 = M_4 = e^2$, Theorems 4.7 and 4.8 give the two-sided estimate

$$C_\beta h \leq E_h^{\mathrm{FO}} \leq B_{\mathrm{FO}}(h), \qquad C_\beta = \frac{e\sinh\pi}{2\sqrt{2}\,(\pi\cosh\pi + \beta\sinh\pi)}.$$

Here $C_0 = 0.3047737605\ldots$ and $C_2 = 0.1864919154\ldots$. At $N = 128$, the Neumann lower bound is 2.3810450×10^{-3} and the upper bound is 1.0128533×10^{-1}, whereas the measured maximum error is 6.8528917×10^{-3}. Thus the upper bound is approximately 14.8 times the measured error. Table 1 also gives the Robin comparison.

The bound is conservative because its proof replaces spatially varying defects by uniform derivative bounds and uses a common comparison barrier for the interior and boundary forcing. It remains a valid error estimate, but should not be read as a sharp prediction of the observed

error magnitude. Likewise, C_β is a guaranteed lower-bound constant, rather than an identified leading coefficient. Together the bounds prove $E_h^{\mathrm{FO}} = \Theta(h)$ for this benchmark and each fixed $\beta \geq 0$. For G, the established conclusion is an $O(h^2)$ upper bound; no matching lower bound is proved here. For affine solutions the defect bounds vanish for both closures, and for quadratic solutions they vanish for G, so an upper-order statement alone does not imply a nonzero error for every smooth solution.

Boundary	C_β	$C_\beta h$	$\widehat{E}_h^{\mathrm{FO}}$	$B_{\mathrm{FO}}(h)$	$B_{\mathrm{FO}}/\widehat{E}_h^{\mathrm{FO}}$
Neumann ($\beta = 0$)	0.304774	2.38105×10^{-3}	6.85289×10^{-3}	1.01285×10^{-1}	14.8
Robin ($\beta = 2$)	0.186492	1.45697×10^{-3}	4.15872×10^{-3}	1.01285×10^{-1}	24.4

Table 1: FO maximum errors at $h = 1/128$: theoretical lower bound, measured error and theoretical upper bound. The theoretical inequalities concern the exact discrete solution. The hatted quantity is evaluated from the floating-point computed solution, using the existing refinement data; the comparison is illustrative and does not replace an algebraic-error analysis.

5 Matrix structure and solver choice

5.1 Weighted symmetry and the energy identity

For either closure define the positive diagonal matrix W_h^c by

$$(W_h^c)_{kk} = \begin{cases} 1, & k \text{ corresponds to } \mathcal{I}_h, \\ 1/h, & k \text{ corresponds to } \mathcal{B}_h,\ c = \mathrm{FO}, \\ 1/2, & k \text{ corresponds to } \mathcal{B}_h,\ c = \mathrm{G}, \end{cases} \qquad K_h^c = W_h^c A_h^c. \tag{5.1}$$

For a vector V, extended by zero on $\mathcal{D}_h$, write

$$\begin{aligned} S_x(V) &= \sum_{j=1}^{N-1} \sum_{i=0}^{N-1} (V_{i+1,j} - V_{i,j})^2, \\ S_{y,I}(V) &= \sum_{i=1}^{N-1} \sum_{j=0}^{N-1} (V_{i,j+1} - V_{i,j})^2, \\ S_{y,B}(V) &= \sum_{j=0}^{N-1} (V_{N,j+1} - V_{N,j})^2. \end{aligned} \tag{5.2}$$

Proposition 5.1 (Symmetry, energy identity and positive definiteness)**.** *For $N \geq 2$ and $\beta \geq 0$, K_h^c is real symmetric positive definite. For every real vector $\boldsymbol{V}$,*

$$h^2 \boldsymbol{V}^T K_h^c \boldsymbol{V} = S_x(V) + S_{y,I}(V) + \theta_c S_{y,B}(V) + \beta h \sum_{j=1}^{N-1} V_{N,j}^2, \quad \theta_{\mathrm{FO}} = 0, \quad \theta_{\mathrm{G}} = \tfrac{1}{2}. \tag{5.3}$$

Proof. Interior–interior off-diagonal pairs already agree. On each interior–boundary pair, multiplying the boundary coefficient by $1/h$ (FO) or $1/2$ (G) makes both coefficients $-1/h^2$. In G the two tangential boundary coefficients both become $-1/(2h^2)$; in FO both are zero. There are no other off-diagonal pairs, so K_h^c is symmetric.

For a fixed j, abbreviate $v_i = V_{i,j}$, with $v_0 = 0$. Expansion of the squares proves the finite summation identity

$$\sum_{i=1}^{N-1} v_i(2v_i - v_{i-1} - v_{i+1}) + v_N(v_N - v_{N-1}) = \sum_{i=0}^{N-1} (v_{i+1} - v_i)^2. \tag{5.4}$$

Indeed both sides equal $2\sum_{i=1}^{N-1} v_i^2 + v_N^2 - 2\sum_{i=1}^{N-1} v_i v_{i+1}$. For a vertical sequence z_j with $z_0 = z_N = 0$, the same expansion gives

$$\sum_{j=1}^{N-1} z_j(2z_j - z_{j-1} - z_{j+1}) = \sum_{j=0}^{N-1} (z_{j+1} - z_j)^2. \tag{5.5}$$

Multiplication of the scaled matrix by h^2 leaves the usual interior rows and the following boundary rows:

$$\begin{cases} V_{N,j} - V_{N-1,j} + \beta h V_{N,j}, & c = \mathrm{FO}, \\ V_{N,j} - V_{N-1,j} + \frac{1}{2}(2V_{N,j} - V_{N,j-1} - V_{N,j+1}) + \beta h V_{N,j}, & c = \mathrm{G}. \end{cases}$$

Apply (5.4) to every horizontal sequence and (5.5) to each interior column. In G also apply the vertical identity to the right column with factor $1/2$. Summing the remaining diagonal Robin terms proves (5.3), including the boundary edges adjoining the Dirichlet corners.

Every term on the right of (5.3) is nonnegative. If the quadratic form is zero, $S_x(V) = 0$, so every horizontal difference is zero. Starting from $V_{0,j} = 0$ then gives $V_{i,j} = 0$ for all unknowns. The symmetric quadratic form is therefore strictly positive for every nonzero vector, proving positive definiteness. □

It follows that

$$\|V\|_{E,h,c} = \left(h^2 \boldsymbol{V}^T K_h^c \boldsymbol{V}\right)^{1/2} \tag{5.6}$$

is a norm on the unknown vectors with zero Dirichlet extension.

5.2 Discrete coercivity and energy-error bounds

Lemma 5.2 (Discrete Poincaré and right-edge trace bounds)**.** *For $V = 0$ on $\mathcal{D}_h$, $N \geq 2$ and $\beta \geq 0$,*

$$\|V\|_{2,h}^2 \leq S_x(V) \leq \|V\|_{E,h,c}^2, \qquad \|V_B\|_{\Gamma,h}^2 \leq S_x(V) \leq \|V\|_{E,h,c}^2. \tag{5.7}$$

Proof. For fixed j, put $d_{r,j} = V_{r+1,j} - V_{r,j}$. Since $V_{0,j} = 0$, $V_{i,j} = \sum_{r=0}^{i-1} d_{r,j}$. The Cauchy–Schwarz inequality gives

$$|V_{i,j}|^2 \leq i \sum_{r=0}^{i-1} d_{r,j}^2 \leq N \sum_{r=0}^{N-1} d_{r,j}^2.$$

Using the zero Dirichlet values in (3.3), and $hN = 1$,

$$\|V\|_{2,h}^2 = h^2 \sum_{j=1}^{N-1} \left(\sum_{i=1}^{N-1} V_{i,j}^2 + \tfrac{1}{2} V_{N,j}^2 \right) \leq h^2 N^2 S_x(V) = S_x(V).$$

The same pointwise bound with $i = N$ yields $\|V_B\|_{\Gamma,h}^2 = h\sum_j V_{N,j}^2 \leq hNS_x(V) = S_x(V)$. Finally (5.3) bounds $S_x(V)$ by the energy. □

Proposition 5.3 (Uniform energy stability)**.** *Under the hypotheses of Proposition 4.6,*

$$\|V\|_{E,h,c} \leq \|F\|_{I,h} + \gamma_c \|S\|_{\Gamma,h}, \qquad \gamma_{\mathrm{FO}} = 1, \quad \gamma_{\mathrm{G}} = h/2. \tag{5.8}$$

Proof. Let w_c be the boundary weight in (5.1), so that $\gamma_c = h w_c$. The equation $A_h^c \boldsymbol{V} = \boldsymbol{d}$, where $\boldsymbol{d}$ has entries F and S, gives

$$\|V\|_{E,h,c}^2 = h^2 \sum_{\mathcal{I}_h} V_{i,j} F_{i,j} + h^2 w_c \sum_{j=1}^{N-1} V_{N,j} S_j.$$

Cauchy–Schwarz and Lemma 5.2 imply

$$\begin{aligned}\|V\|_{E,h,c}^2 &\le \|V_I\|_{I,h}\|F\|_{I,h} + hw_c\|V_B\|_{\Gamma,h}\|S\|_{\Gamma,h}\\ &\le \|V\|_{E,h,c}(\|F\|_{I,h} + \gamma_c\|S\|_{\Gamma,h}).\end{aligned}$$

Here $\|V_I\|_{I,h} \le \|V\|_{2,h} \le \|V\|_{E,h,c}$. If the energy norm is zero the claim is immediate; otherwise division by it proves (5.8). □

Corollary 5.4 (Energy and discrete L^2 convergence)**.** *Under Assumption 2.1, the discretisation errors satisfy*

$$\|e_h^{\mathrm{FO}}\|_{2,h} \le \|e_h^{\mathrm{FO}}\|_{E,h,\mathrm{FO}} \le \tfrac{M_4}{6}h^2 + \tfrac{M_2}{2}h, \tag{5.9}$$

$$\|e_h^{\mathrm{G}}\|_{2,h} \le \|e_h^{\mathrm{G}}\|_{E,h,\mathrm{G}} \le \tfrac{M_4+M_3}{6}h^2 + \tfrac{M_4}{24}h^3. \tag{5.10}$$

Thus both norms have upper orders one (FO) and two (G).

Proof. Apply (5.8) to the error equation with $F = -\tau_I$ and $S = -\tau_B^c$. Since $h^2(N-1)^2 \le 1$ and $h(N-1) \le 1$, the restricted norms satisfy $\|\tau_I\|_{I,h} \le \|\tau_I\|_{\infty,I}$ and $\|\tau_B^c\|_{\Gamma,h} \le \|\tau_B^c\|_{\infty,B}$. Insert (4.6). The boundary factors are 1 for FO and $h/2$ for G, giving the two energy bounds exactly as stated. Lemma 5.2 supplies the L^2 bounds. □

The energy norm depends on the selected boundary closure. It provides a further theoretical and numerical check, while the common nodal maximum norm remains the primary target for comparing computational cost.

5.3 Solver implications of the proved matrix structure

The raw matrices are nonsymmetric and invertible. We use a sparse direct baseline through SciPy's `spsolve` interface [6], with CSC input and `use_umfpack=False`, explicitly fixing the SuperLU backend. Each call includes a fresh factorisation; no factor is reused between timed repetitions. Multiplication by the invertible positive W_h^c gives the equivalent system $K_h^c \boldsymbol{U}_h^c = W_h^c \boldsymbol{b}_h^c$. Proposition 5.1 proves the symmetry and positive definiteness required for CG [7]. Its diagonal is positive because $(K_h^c)_{kk} = \boldsymbol{e}_k^T K_h^c \boldsymbol{e}_k > 0$. Consequently, the Jacobi inverse $\mathrm{diag}(K_h^c)^{-1}$ is an admissible positive definite preconditioner. The baseline and cross-checks below test these solver choices; the structural proof alone makes no runtime claim.

6 Verification

The conclusions of this report rest on a chain of evidence rather than on a single number. This section states which claims are made, which checks support each of them, and then reports the checks that concern the assembled equations and the linear solves. The convergence evidence itself is reported in Section 7.

6.1 Evidence plan

Table 2 lists each main claim together with the independent checks that support it. A check is counted as independent only if it does not share a derivation or an implementation with the code under test; agreement between two runs of the same code is not used as evidence.

6.2 Equation-level checks

The following checks test whether the assembled matrix discretises the intended equations. None of them involves an iterative solver.

Claim	Supporting evidence	Where
FO maximum error is $\Theta(h)$	Theorems 4.7 and 4.8; fitted orders on $N = 128, 256, 512$; FDM–FEM difference decays at first order	Sections 7.2 and 6.3
G error is $O(h^2)$	Theorem 4.7 and Corollary 5.4; fitted orders in the nodal maximum and discrete L^2 norms; FDM–FEM difference decays at second order	Sections 7.2 and 6.3
The assembled rows are the intended equations	Hand-derived boundary row; polynomial defect identities; separately assembled uneliminated system; independent P1 finite-element solve	Sections 6.2 and 6.3
Linear-solver error is negligible	Direct-solve residual; ILU- and Jacobi-preconditioned GMRES tolerance sweeps; weighted CG	Section 6.4
The weighted matrices are symmetric positive definite	Proposition 5.1; edge-sum identity and smallest-eigenvalue estimate at $N = 32$	Sections 5 and 7.4
G is cheaper at a fixed nodal accuracy	Timing protocol with warm-up, five samples and fixed error targets	Section 7.5

Table 2: Claims of this report and the checks that support them.

Hand-derived boundary row. At $N = 4$, $h = 1/4$, $j = 1$, the ghost row has diagonal $64+8\beta$, west coefficient -32 and retained north coefficient -16. After eliminating the bottom-right corner, its right-hand side is $(6 + 8\beta)e^{1.25} + 16e$. These values check the inward factor, outward-normal sign and corner contribution separately.

Manufactured functions with different polynomial structure. For $u = 1 + x + y$, both closures have zero defect in exact arithmetic. For $u = 1 + x^2 + y^2$, the ghost defect is zero and the FO boundary defect is $-h$. An additional check uses

$$u = 1 + x^3 + y^4, \qquad f = -6x - 12y^2, \qquad s_\beta = 3 + \beta(2 + y^4).$$

Direct polynomial substitution predicts

$$\tau_I = -2h^2, \qquad \tau_B^{\mathrm{FO}} = -3h + h^2, \qquad \tau_B^{\mathrm{G}} = 2h - 2h^2.$$

These identities are checked for both boundary variants at $N = 2, 3, 4, 6$, using data derived without applying the discrete matrix. They exercise nonzero forcing as well as boundary derivatives.

A separately assembled uneliminated system. On the same four small grids, an additional dense system retains every Dirichlet variable and, for G, every ghost variable. It imposes explicit Dirichlet identity equations, the original five-point PDE and centred boundary equations. Its physical-node solution is compared with the reduced sparse solve in all sixteen configurations. Agreement is required within absolute and relative tolerances 10^{-11}. This checks elimination and assembly through a different system layout.

Weighted-structure checks. For the energy identity, a seeded random vector is tested against the independent edge sums, with relative discrepancy below 10^{-12}. At $N = 32$, `eigsh` with `which='SA'` and tolerance 10^{-10} estimates the smallest eigenvalue [10]. This is a finite numerical corroboration of Proposition 5.1, not its proof. The results are reported in Section 7.4.

What these checks do not show. A dense solve of the same reduced matrix, or the iterative solves of Section 6.4, test linear algebra only. Agreement alone would not establish that the matrix discretises the intended PDE. The hand row, the independent data and the uneliminated construction supply the complementary equation checks, and the independent finite-element solve below supplies a check that shares neither derivation nor code with the finite-difference stencils.

6.3 Independent finite-element oracle

An independent P1 triangular finite-element solve uses the weak form (2.7)–(2.8): diffusion in Ω, Robin mass on Γ_B when $\beta = 2$, volume load f, boundary load s_β, and exact Dirichlet values on the other three edges. It is assembled by `scikit-fem` version 11.0.0 [12] rather than by the finite-difference stencil code. We compare FEM and both FDM closures only at their common Cartesian vertices; no interpolation or norm-conversion claim is required.

Table 3 shows the independent FEM error against e^{x+y} and the FDM–FEM difference. The latter decays at roughly first order for FO and second order for G, consistent with the FDM refinement evidence. Agreement between two discretisations of the same weak problem is a cross-check, not a proof that either is exact.

Boundary	Closure	N	FEM error	FDM error	FDM–FEM difference
Neumann	FO	16	2.29×10^{-3}	5.83×10^{-2}	5.60×10^{-2}
Neumann	G	16	2.29×10^{-3}	8.61×10^{-4}	3.15×10^{-3}
Neumann	FO	32	5.74×10^{-4}	2.81×10^{-2}	2.76×10^{-2}
Neumann	G	32	5.74×10^{-4}	2.16×10^{-4}	7.90×10^{-4}
Neumann	FO	64	1.44×10^{-4}	1.38×10^{-2}	1.37×10^{-2}
Neumann	G	64	1.44×10^{-4}	5.41×10^{-5}	1.98×10^{-4}
Neumann	FO	128	3.59×10^{-5}	6.85×10^{-3}	6.82×10^{-3}
Neumann	G	128	3.59×10^{-5}	1.35×10^{-5}	4.94×10^{-5}
Robin	FO	16	7.21×10^{-4}	3.43×10^{-2}	3.36×10^{-2}
Robin	G	16	7.21×10^{-4}	5.31×10^{-4}	1.25×10^{-3}
Robin	FO	32	1.81×10^{-4}	1.69×10^{-2}	1.67×10^{-2}
Robin	G	32	1.81×10^{-4}	1.33×10^{-4}	3.14×10^{-4}
Robin	FO	64	4.53×10^{-5}	8.36×10^{-3}	8.31×10^{-3}
Robin	G	64	4.53×10^{-5}	3.34×10^{-5}	7.86×10^{-5}
Robin	FO	128	1.13×10^{-5}	4.16×10^{-3}	4.15×10^{-3}
Robin	G	128	1.13×10^{-5}	8.34×10^{-6}	1.97×10^{-5}

Table 3: Independent P1 triangular FEM solution of the weak form in Section 2, compared at common grid vertices. Errors are nodal maxima against the manufactured exact solution; the difference is not itself an error bound.

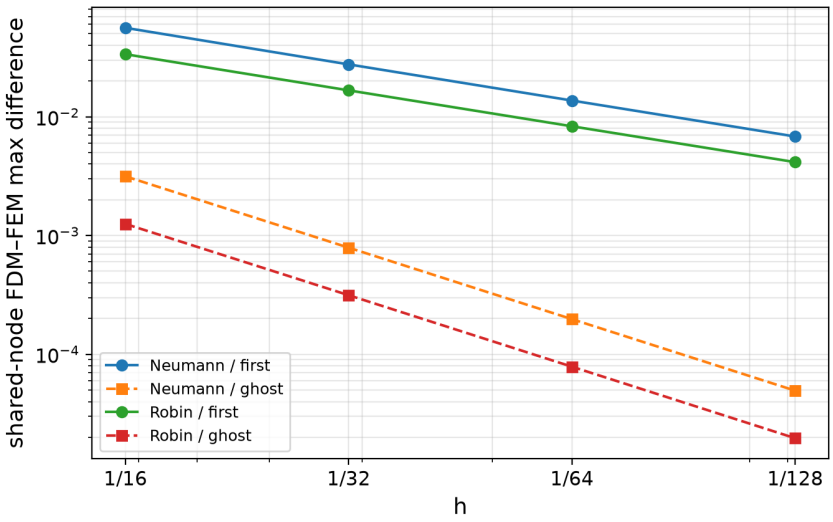


Figure 2: Maximum FDM–FEM difference at shared vertices. The FEM code assembles the continuum weak form independently of the FDM rows.

6.4 Separation of algebraic and discretisation error

Residual measures. For a computed vector $\widetilde{\boldsymbol{U}}$, set $\boldsymbol{r} = \boldsymbol{b} - A\widetilde{\boldsymbol{U}}$ and report

$$\eta_{\mathrm{cw}} = \max_k \frac{|r_k|}{(|A||\widetilde{\boldsymbol{U}}| + |\boldsymbol{b}|)_k}, \qquad \eta_{A,\infty} = \frac{\|\boldsymbol{r}\|_\infty}{\|A\|_\infty \|\widetilde{\boldsymbol{U}}\|_\infty + \|\boldsymbol{b}\|_\infty}. \tag{6.1}$$

A zero denominator with zero numerator contributes zero. The componentwise measure is unchanged by positive diagonal row scaling, so it compares the different raw row scales meaningfully. Neither residual is a discretisation error or a forward-error bound. The baseline is a sparse direct solve through SciPy's `spsolve` interface [6] with CSC input and `use_umfpack=False`, which fixes the SuperLU backend. Each call includes a fresh factorisation, and the solver has no user-set stopping tolerance. Its maximum componentwise residual over the twenty-four refinement runs is 2.12×10^{-12}.

Acceptance criterion. The iterative solves below are cross-checks on a representative mesh. Each is accepted when

$$\rho_{\mathrm{alg}} = \frac{\|U_{\mathrm{iterative}} - U_{\mathrm{direct}}\|_\infty}{\|U_{\mathrm{direct}} - u_h^*\|_\infty} < 10^{-2}. \tag{6.2}$$

The boundary-mean quantity $Q_h = (N-1)^{-1} \sum_{j=1}^{N-1} U_{N,j}$ is monitored as well. Agreement at tighter tolerances supports negligible solve error on this representative grid; it is not a general proof for arbitrary systems.

ILU-preconditioned GMRES. At $N = 64$, we cross-check the direct solution using GMRES [8]. Let $D_{kk} = (\sum_j |a_{kj}|)^{-1}$; GMRES solves $DAU = Db$ with ILU [9], drop tolerance 10^{-4}, fill factor 20, restart 50, at most 1000 restart cycles, zero initial vector, zero absolute tolerance and relative tolerances $10^{-8}, 10^{-10}, 10^{-12}$. The residual stopping test uses the scaled system's Euclidean norm; the original-system diagnostics in (6.1) are evaluated separately. D is not the analytical symmetriser W_h^c.

Table 4 implements this tolerance sweep. Every run returns status zero. Even at rtol 10^{-8} the largest ratio in (6.2) is 3.67×10^{-5}, below 10^{-2}; at 10^{-12} it is 4.30×10^{-8}. At the tightest tolerance, the largest change in the boundary-mean quantity relative to the direct solution is 1.75×10^{-12}.

The residual, solution difference and quantity of interest thus provide separate evidence for the linear solve. Small nonmonotone ρ_{alg} changes between 10^{-10} and 10^{-12} reflect the floating-point difference floor between the two solutions; they are not interpreted as a discretisation effect.

Boundary	Closure	rtol	Iterations	η_{cw}	ρ_{alg}
Neumann	FO	1×10^{-8}	4	5.92×10^{-11}	6.69×10^{-8}
Neumann	FO	1×10^{-10}	5	1.37×10^{-13}	2.55×10^{-10}
Neumann	FO	1×10^{-12}	5	9.59×10^{-14}	2.91×10^{-10}
Neumann	G	1×10^{-8}	4	7.09×10^{-11}	2.20×10^{-5}
Neumann	G	1×10^{-10}	5	1.95×10^{-13}	8.92×10^{-8}
Neumann	G	1×10^{-12}	5	6.52×10^{-14}	3.55×10^{-8}
Robin	FO	1×10^{-8}	4	6.48×10^{-11}	1.17×10^{-7}
Robin	FO	1×10^{-10}	5	2.14×10^{-13}	5.19×10^{-10}
Robin	FO	1×10^{-12}	5	1.20×10^{-13}	3.66×10^{-10}
Robin	G	1×10^{-8}	4	9.19×10^{-11}	3.67×10^{-5}
Robin	G	1×10^{-10}	5	1.21×10^{-13}	1.07×10^{-7}
Robin	G	1×10^{-12}	5	9.91×10^{-14}	4.30×10^{-8}

Table 4: GMRES–ILU cross-check at $N = 64$, with atol $= 0$. Residuals are evaluated on the original raw system. The ratio ρ_{alg} is the iterative/direct maximum difference divided by the measured direct-solution error; the acceptance target is 10^{-2}.

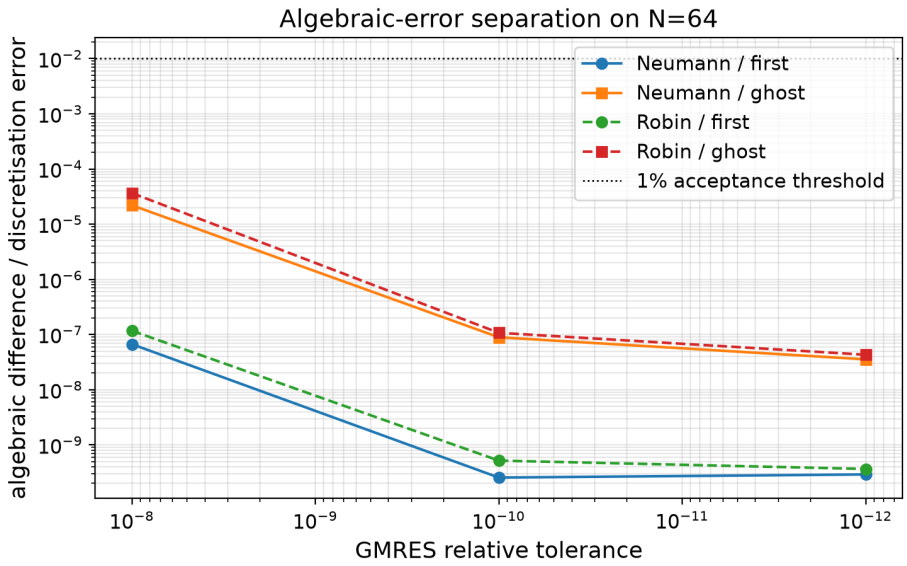


Figure 3: GMRES/direct differences relative to discretisation errors at $N = 64$. All ratios lie below the one-percent acceptance line. The flattening at tight tolerances is consistent with a solve-difference floor and is not interpreted as a new discretisation rate.

A weaker, genuinely iterative check. ILU-preconditioned GMRES converges in only a few iterations and is a relatively weak independence check. Table 5 therefore uses GMRES on the original matrix with only a Jacobi (diagonal) preconditioner and no approximate factorisation: zero initial vector, zero absolute tolerance, restart 50, at most 1000 restart cycles and relative tolerances 10^{-8} and 10^{-10}. Iterations are counted from the residual-norm callback. At 10^{-8} the algebraic/discretisation ratios for the G closures exceed one percent; tightening to 10^{-10} reduces all four ratios below this threshold. This distinction prevents the weaker preconditioner from

being credited with accuracy it did not achieve, and shows that a stopping tolerance must be chosen relative to the discretisation error of the closure being solved.

Boundary	Closure	rtol	Iterations	η_{cw}	ρ_{alg}
Neumann	FO	1×10^{-8}	452	7.19×10^{-10}	3.14×10^{-4}
Neumann	FO	1×10^{-10}	588	1.23×10^{-11}	1.99×10^{-6}
Neumann	G	1×10^{-8}	435	9.29×10^{-10}	7.40×10^{-2}
Neumann	G	1×10^{-10}	582	1.23×10^{-11}	5.47×10^{-4}
Robin	FO	1×10^{-8}	392	1.11×10^{-9}	2.46×10^{-4}
Robin	FO	1×10^{-10}	518	1.31×10^{-11}	2.90×10^{-6}
Robin	G	1×10^{-8}	393	1.10×10^{-9}	6.14×10^{-2}
Robin	G	1×10^{-10}	519	1.13×10^{-11}	7.47×10^{-4}

Table 5: Jacobi-preconditioned GMRES on the original matrix at $N = 64$, without ILU factorisation.

Weighted CG. Weighted CG uses $K_h^c U = W_h^c b$, Jacobi preconditioning, zero initial vector, rtol 10^{-10}, atol zero and at most 10000 iterations; CG applies because Proposition 5.1 proves the weighted matrix symmetric positive definite, and the Jacobi inverse $\mathrm{diag}(K_h^c)^{-1}$ is an admissible positive definite preconditioner [7]. The reported normwise weighted residual is

$$\eta_{K,\infty} = \frac{\|Wb - K\widetilde{U}\|_\infty}{\|K\|_\infty \|\widetilde{U}\|_\infty + \|Wb\|_\infty}.$$

Table 6 also gives the original-system η_{cw}, so differently scaled residuals are not conflated. All four CG solves return status zero and require 280–289 iterations. The largest CG/direct difference ratio is 3.06×10^{-5}. This verifies the implementation of the proved eligible system. It does not establish that CG is faster than direct factorisation.

Boundary	Closure	Iterations	$\eta_{K,\infty}$	η_{cw}	ρ_{alg}
Neumann	FO	289	4.30×10^{-12}	1.98×10^{-11}	6.82×10^{-8}
Neumann	G	287	3.79×10^{-12}	1.66×10^{-11}	1.63×10^{-5}
Robin	FO	280	5.27×10^{-12}	2.57×10^{-11}	8.97×10^{-8}
Robin	G	280	3.84×10^{-12}	1.87×10^{-11}	3.06×10^{-5}

Table 6: Jacobi-preconditioned CG on $KU = Wb$ at $N = 64$, rtol 10^{-10}, atol zero. All return status zero. The two residual columns distinguish weighted normwise and original componentwise measures.

Scope. These comparisons use independent linear solvers for the same system; they do not replace the equation-level checks of Section 6.2. All iterative exit codes and residuals are recorded. The tolerance sweep is conducted on the required representative mesh rather than claimed as a forward-error proof for every possible refinement.

6.5 Reproducibility and test suite

The script `code/run_all.py` executes the four configurations, the six-mesh study, the fixed-target cost study and the solver cross-checks. It generates CSV data in `data/`, LaTeX tables in `generated/` and figures in `figures/`, so that no numerical value in the report is entered by hand. The assembly, solve and diagnostics live in `code/models.py`; the experiments in `code/experiments.py`; the independent finite-element solve in `code/fem_oracle.py`; and the pure-Neumann extension of Section 7.6 in `code/pure_neumann.py`. The pure-Neumann code

treats all edges as flux boundaries, computes left- and right-null diagnostics and solves a bordered mean-zero system; it is not substituted for the mixed-boundary model.

The executable test suite contains thirteen test methods. It checks dimensions, signs, finite entries, invalid options, corner ownership, exact rows, manufactured defects, observed rates, the weighted energy identity, the finite-element oracle, the pure-Neumann null space and gauge, and the agreement of the ghost-closure compatibility defect with a trapezoidal quadrature.

The bundled data were generated on Windows 11 with Python 3.12.14, NumPy 2.5.4, SciPy 1.18.1 and Matplotlib 3.11.2. The version and timing settings are recorded in `data/run_metadata.json`. Reproduction on another machine may change timings and the last few residual digits.

7 Numerical experiments

This section reports the main experiments. Sections 7.1 to 7.4 compare the two closures through convergence, error location and matrix structure; Section 7.5 compares their measured cost at fixed accuracy; Section 7.6 is a separate extension. The checks that establish that the numbers can be trusted are collected in Section 6.

7.1 Protocol, norms and order fitting

Both boundary variants and both closures use the same data, ordering and Dirichlet ownership. The required refinement family is $N = 16, 32, 64, 128$, extended to $N = 256, 512$ for finer-grid validation. For a computed full-grid error $e = \widetilde{U} - u_h^*$, we report $E_{2,h} = \|e\|_{2,h}$ and $E_{\infty,h} = \|e\|_{\infty,h}$ using exactly (3.3). At consecutive levels,

$$p_m(N) = \frac{\log(E_{m,2h}/E_{m,h})}{\log 2}, \qquad m = 2, \infty.$$

After inspecting pairwise trends, we fit $\log E = p \log h + C$ on $N = 128, 256, 512$. We also report the Richardson difference estimate $p_R = \log_2[(E_{128} - E_{256})/(E_{256} - E_{512})]$ when defined. Three-point fits alone do not prove an asymptotic expansion. Software versions, timing settings and the way the data are generated are described in Section 6.5.

7.2 Global convergence of the two closures

Boundary	Closure	N	h	$E_{2,h}$	p_2	$E_{\infty,h}$	p_∞	η_{cw}
Neumann	FO	16	1/16	1.786×10^{-2}	–	5.832×10^{-2}	–	1.95×10^{-14}
Neumann	FO	32	1/32	8.515×10^{-3}	1.069	2.815×10^{-2}	1.051	2.75×10^{-14}
Neumann	FO	64	1/64	4.161×10^{-3}	1.033	1.383×10^{-2}	1.025	7.32×10^{-14}
Neumann	FO	128	1/128	2.058×10^{-3}	1.016	6.853×10^{-3}	1.013	2.17×10^{-13}
Neumann	FO	256	1/256	1.023×10^{-3}	1.008	3.411×10^{-3}	1.007	9.46×10^{-13}
Neumann	FO	512	1/512	5.102×10^{-4}	1.004	1.702×10^{-3}	1.003	1.19×10^{-12}
Neumann	G	16	1/16	2.206×10^{-4}	–	8.611×10^{-4}	–	2.41×10^{-16}
Neumann	G	32	1/32	5.499×10^{-5}	2.004	2.163×10^{-4}	1.993	3.78×10^{-16}
Neumann	G	64	1/64	1.374×10^{-5}	2.001	5.412×10^{-5}	1.998	3.37×10^{-16}
Neumann	G	128	1/128	3.434×10^{-6}	2.000	1.354×10^{-5}	1.999	4.44×10^{-16}
Neumann	G	256	1/256	8.584×10^{-7}	2.000	3.385×10^{-6}	2.000	7.15×10^{-16}
Neumann	G	512	1/512	2.146×10^{-7}	2.000	8.461×10^{-7}	2.000	7.46×10^{-16}
Robin	FO	16	1/16	1.059×10^{-2}	–	3.430×10^{-2}	–	4.89×10^{-15}
Robin	FO	32	1/32	5.143×10^{-3}	1.042	1.685×10^{-2}	1.026	3.50×10^{-14}
Robin	FO	64	1/64	2.536×10^{-3}	1.020	8.355×10^{-3}	1.012	9.16×10^{-14}
Robin	FO	128	1/128	1.259×10^{-3}	1.010	4.159×10^{-3}	1.007	2.39×10^{-13}
Robin	FO	256	1/256	6.276×10^{-4}	1.005	2.075×10^{-3}	1.003	1.48×10^{-12}
Robin	FO	512	1/512	3.133×10^{-4}	1.002	1.036×10^{-3}	1.002	2.12×10^{-12}
Robin	G	16	1/16	1.273×10^{-4}	–	5.307×10^{-4}	–	2.27×10^{-16}
Robin	G	32	1/32	3.162×10^{-5}	2.009	1.332×10^{-4}	1.994	2.81×10^{-16}
Robin	G	64	1/64	7.892×10^{-6}	2.003	3.335×10^{-5}	1.998	4.60×10^{-16}
Robin	G	128	1/128	1.972×10^{-6}	2.001	8.340×10^{-6}	2.000	4.16×10^{-16}
Robin	G	256	1/256	4.930×10^{-7}	2.000	2.085×10^{-6}	2.000	8.80×10^{-16}
Robin	G	512	1/512	1.232×10^{-7}	2.000	5.213×10^{-7}	2.000	9.81×10^{-16}

Table 7: Global refinement errors for both boundary data and both closures; pairwise orders use adjacent grids.

Table 7 shows decreasing errors on every refinement. FO pairwise orders approach one from above in both global norms, while G orders are close to two. The maximum componentwise direct-solve residual across the twenty-four runs in Table 7 is 2.12×10^{-12}. Its small size is supplemented by the independent solve checks of Section 6.4; it is not used alone to establish correct boundary discretisation.

Boundary	Closure	p_2	p_∞	p_B	p_I	$p_{R,2}$	$p_{R,\infty}$
Neumann	FO	1.006	1.005	1.005	0.993	1.012	1.010
Neumann	G	2.000	2.000	2.000	1.984	2.000	2.000
Robin	FO	1.004	1.003	1.003	0.990	1.007	1.005
Robin	G	2.000	2.000	2.000	1.980	2.000	2.000

Table 8: Least-squares slopes and Richardson difference-based estimates on $N = 128, 256, 512$. Three points do not independently establish an asymptotic regime.

Table 8 distinguishes a theorem's upper order from a measured slope. The FO maximum-error rate is also supported by the independent lower bound in Theorem 4.8. For G, observed second order agrees with the proved upper bound; no mathematical lower bound for its error is inferred from these meshes.

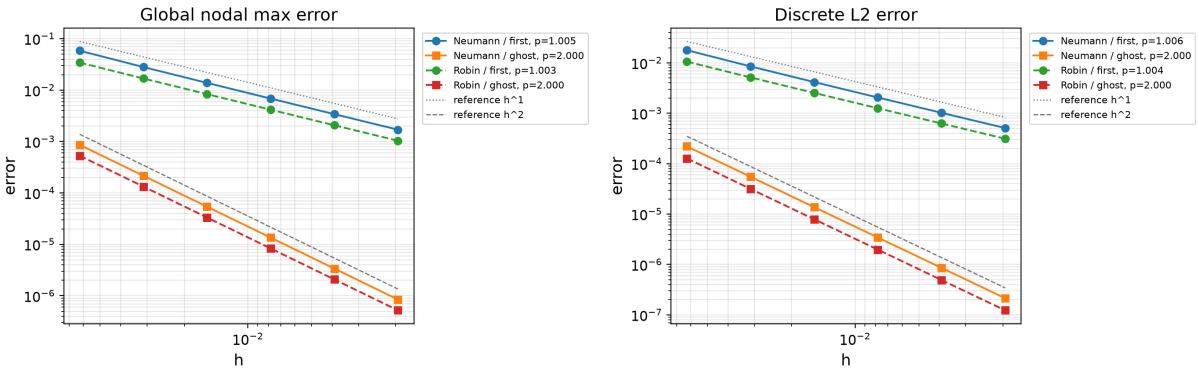


Figure 4: Global maximum and trapezoidal discrete L^2 refinement errors. Legends give the fitted slopes on $N = 128, 256, 512$; dotted/dashed reference curves have slopes one and two and arbitrary vertical placement. Errors approach the orders predicted by Theorem 4.7 and Corollary 5.4.

7.3 Boundary error, interior error and finite-grid deviations

We evaluate $E_{\infty,B} = \max_{1\le j<N} |e_{N,j}|$ and $E_{\infty,I} = \max_{(i,j)\in\mathcal{I}_h} |e_{i,j}|$ separately. The CSV data also retain the restricted L^2 errors defined by (3.4). This classifies errors by where they are measured. It is not an additive separation of the effects of boundary and interior truncation defects, which are coupled by the elliptic solve.

Boundary	Closure	N	$E_{\infty,B}$	p_B	$E_{\infty,I}$	p_I
Neumann	FO	16	5.832×10^{-2}	–	4.863×10^{-2}	–
Neumann	FO	32	2.815×10^{-2}	1.051	2.574×10^{-2}	0.918
Neumann	FO	64	1.383×10^{-2}	1.025	1.322×10^{-2}	0.961
Neumann	FO	128	6.853×10^{-3}	1.013	6.701×10^{-3}	0.981
Neumann	FO	256	3.411×10^{-3}	1.007	3.373×10^{-3}	0.990
Neumann	FO	512	1.702×10^{-3}	1.003	1.692×10^{-3}	0.995
Neumann	G	16	8.611×10^{-4}	–	6.706×10^{-4}	–
Neumann	G	32	2.163×10^{-4}	1.993	1.914×10^{-4}	1.809
Neumann	G	64	5.412×10^{-5}	1.998	5.096×10^{-5}	1.909
Neumann	G	128	1.354×10^{-5}	1.999	1.314×10^{-5}	1.956
Neumann	G	256	3.385×10^{-6}	2.000	3.334×10^{-6}	1.978
Neumann	G	512	8.461×10^{-7}	2.000	8.398×10^{-7}	1.989
Robin	FO	16	3.430×10^{-2}	–	2.854×10^{-2}	–
Robin	FO	32	1.685×10^{-2}	1.026	1.537×10^{-2}	0.893
Robin	FO	64	8.355×10^{-3}	1.012	7.978×10^{-3}	0.946
Robin	FO	128	4.159×10^{-3}	1.007	4.064×10^{-3}	0.973
Robin	FO	256	2.075×10^{-3}	1.003	2.051×10^{-3}	0.987
Robin	FO	512	1.036×10^{-3}	1.002	1.030×10^{-3}	0.993
Robin	G	16	5.307×10^{-4}	–	3.928×10^{-4}	–
Robin	G	32	1.332×10^{-4}	1.994	1.149×10^{-4}	1.774
Robin	G	64	3.335×10^{-5}	1.998	3.098×10^{-5}	1.890
Robin	G	128	8.340×10^{-6}	2.000	8.039×10^{-6}	1.946
Robin	G	256	2.085×10^{-6}	2.000	2.047×10^{-6}	1.973
Robin	G	512	5.213×10^{-7}	2.000	5.165×10^{-7}	1.987

Table 9: Spatially restricted maximum errors; this is not an additive source decomposition.



Figure 5: Error maxima on the open right edge and in the interior. The displayed slopes use the same finest-three-level fits. FO boundary error limits global accuracy, and the associated error extends into the interior. The spatial classification does not attribute interior error exclusively to the interior stencil.

For all tested configurations the global maximum equals the right-edge maximum, and the interior maximum is smaller but of comparable size. The interior FO fits approach one, consistent with a boundary defect affecting the full field. Thus a second-order interior stencil does not give a second-order global solution under the FO closure.

The G interior maximum fits are 1.984 and 1.980, slightly below the global fits of two. The pairwise orders in Table 9 increase monotonically towards two as the grid is refined, from about 1.8 at $N = 32$ to about 1.99 at $N = 512$ for both boundary variants. An additional location diagnostic, `data/error_locations.csv`, finds the fine-grid interior maxima on column $i = N-1$. This sampled column moves towards $x = 1$ as h decreases. The interior/global error ratios also move towards one. These observations indicate finite-grid and maximising-location effects; they do not identify a separate asymptotic leading term. The measured deviations are compatible with the uniform $O(h^2)$ upper bound and are not evidence of tolerance contamination, given the much smaller solver differences reported in Section 6.4.

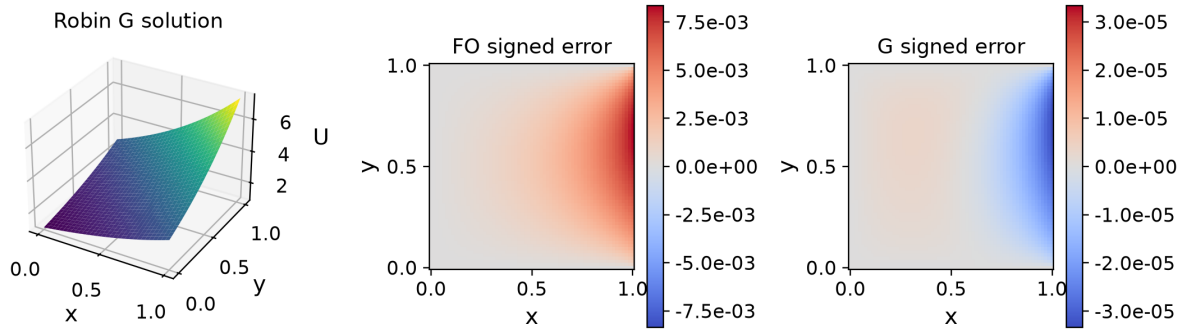


Figure 6: Representative Robin solution and signed errors at $N = 64$. The surface is the G solution. The FO and G error maps use separate, explicit colour scales centred at zero; their colours alone should not be compared as equal error magnitudes. All three Dirichlet edges and four corners have zero imposed-data error.

7.4 Matrix structure and weighted-energy norm

Raw matrices, weighted structure and sparsity. At $N = 32$, $n_{\mathrm{dof}} = 992$, and the exact counts in Proposition 3.2 give 4774 FO and 4834 G nonzeros. The spy plots show the extra right-edge tangential couplings in G. They do not themselves establish symmetry; the Frobenius diagnostic quantifies it.

Boundary	Closure	δ_A	δ_K	$\lambda_{\min}(K)$	Energy defect
Neumann	FO	0.05515	0	11.6098	0
Neumann	G	0.05595	0	11.9409	1.19×10^{-16}
Robin	FO	0.05515	0	14.7168	1.14×10^{-16}
Robin	G	0.05590	0	14.8745	2.33×10^{-16}

Table 10: Raw and weighted matrix checks at $N = 32$: $\delta_A = \|A - A^T\|_F / \|A\|_F$ and $\delta_K = \|K - K^T\|_F / \|K\|_F$. Eigenvalues are numerical estimates; positive definiteness for all stated grids is proved in Proposition 5.1.

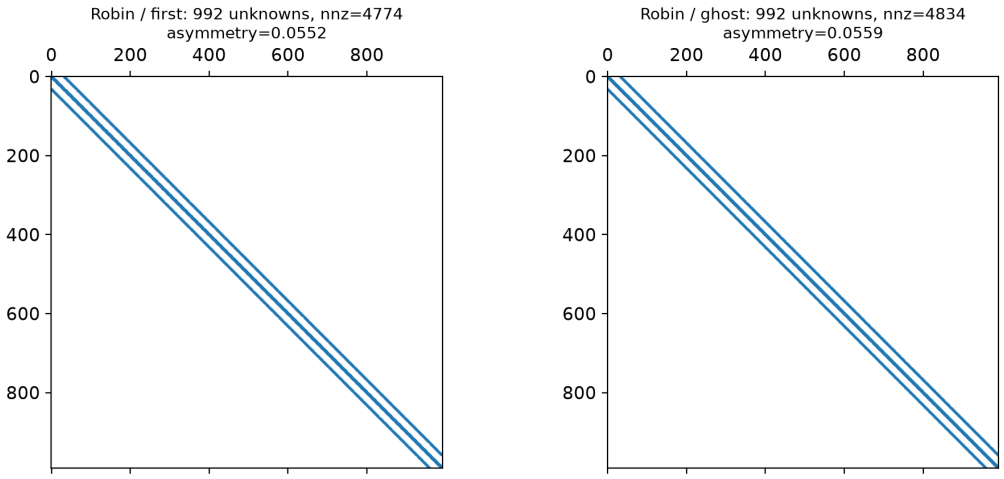


Figure 7: Representative Robin raw matrices at $N = 32$. The sparsity patterns retain nearest-neighbour couplings, while ghost elimination changes the inward coefficient and adds tangential boundary couplings. Both raw matrices are nonsymmetric, as confirmed by the labelled Frobenius ratios.

Table 10 gives nonzero raw asymmetry for all four configurations, exactly zero measured weighted asymmetry on this binary grid, and positive smallest-eigenvalue estimates. The edge-sum identity agrees with the quadratic form at roundoff. These checks corroborate the particular implementation of W_h^c, while Proposition 5.1 supplies the general proof.

Weighted-energy convergence. For the error restricted to unknowns we compute

$$E_{E,h,c} = \left(h^2 \boldsymbol{e}^T K_h^c \boldsymbol{e}\right)^{1/2}.$$

Each quadratic form is independently evaluated through the edge sums. The largest relative discrepancy over the refinement runs is 5.09×10^{-16}. The energy data for every grid used are stored in `data/energy_convergence.csv`.

The energy fits approach one and two, respectively. This additional norm checks the theory beyond nodal errors, but the two closures use different energy norms. The common nodal maximum error, rather than these closure-dependent energies, therefore defines the cost target in Section 7.5.

Boundary	Closure	$E_{E,h}$ at $N = 128$	Fitted p_E	R_E^2
Neumann	FO	9.581×10^{-3}	1.010	0.999997
Neumann	G	2.018×10^{-5}	1.997	1.000000
Robin	FO	7.596×10^{-3}	1.005	1.000000
Robin	G	1.647×10^{-5}	1.996	1.000000

Table 11: Measured energy errors and fine-mesh fits on $N = 32, 64, 128$. The energy norm is closure dependent; it is not the shared cost-comparison norm.

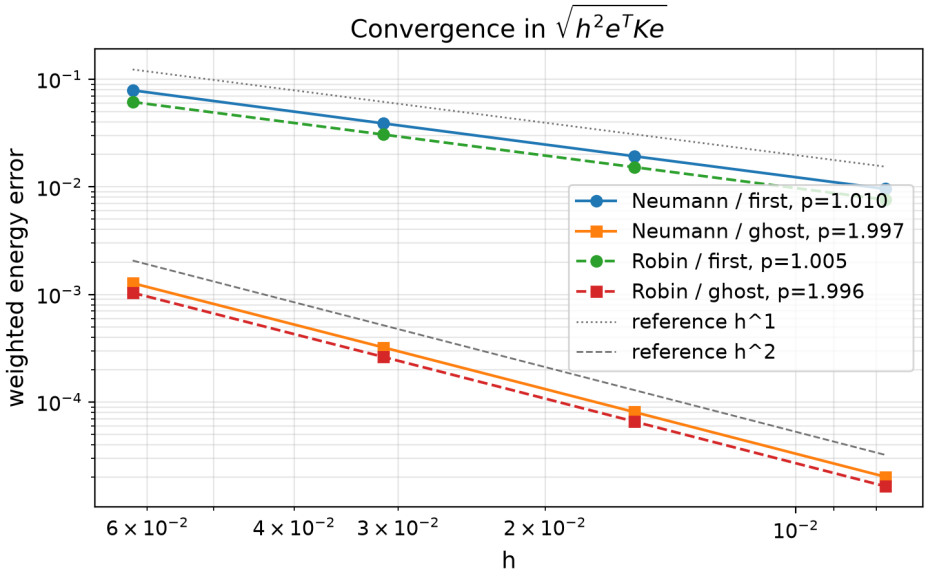


Figure 8: Convergence in the proved closure-dependent energy norm. Reference slopes one and two and fitted slopes on $N = 32, 64, 128$ are shown. FO and G approach the respective upper orders in Corollary 5.4.

7.5 Measured cost at fixed accuracy thresholds

The cost measure is wall-clock time for assembly followed by a fresh sparse direct solve, including COO/CSR/CSC conversion, ordering, factorisation and triangular solves. Imports, a warm-up run, post-solve error/residual diagnostics, independent tests, Krylov and eigenvalue cross-checks, plotting and file I/O are excluded. After one warm-up, five complete totals T_k are measured with `perf_counter`; the reported cost is $\mathrm{median}_k\, T_k$. Separate assembly and solve medians are displayed in Table 12, but their sum is not substituted for the independently measured total median. The sample minimum–maximum remains in the CSV, not a confidence interval. No factorisation or preconditioner is reused.

The fixed targets are $E_{\infty,h} \le 10^{-2}$ and $E_{\infty,h} \le 5 \times 10^{-3}$, for each boundary variant separately. Candidate grids are $N \in \{4, 8, 16, 32, 64, 128, 256, 512\}$; the two extra coarse levels serve this cost study only and do not enter convergence fits. The prescribed selection rule chooses the coarsest tested grid meeting the target. If none does, the target is recorded as unattained; no unmeasured runtime is extrapolated. This is a comparison at a common error bound, not at identical achieved error. In particular, G may oversatisfy a threshold, and the selected grid is not asserted to be a global runtime optimum over all possible N.

Boundary	Target	Closure	N	$E_{\infty,h}$	Assembly (ms)	Solve (ms)	Total (ms)
Neumann	1×10^{-2}	FO	128	6.853×10^{-3}	2.3	28	30
Neumann	1×10^{-2}	G	8	3.377×10^{-3}	0.13	0.084	0.21
Neumann	5×10^{-3}	FO	256	3.411×10^{-3}	9	1.7e+02	1.8e+02
Neumann	5×10^{-3}	G	8	3.377×10^{-3}	0.13	0.084	0.21
Robin	1×10^{-2}	FO	64	8.355×10^{-3}	0.44	5.5	5.9
Robin	1×10^{-2}	G	4	7.326×10^{-3}	0.12	0.043	0.16
Robin	5×10^{-3}	FO	128	4.159×10^{-3}	2.8	33	36
Robin	5×10^{-3}	G	8	2.066×10^{-3}	0.15	0.1	0.25

Table 12: Coarsest successful grid in $\{4, 8, 16, 32, 64, 128, 256, 512\}$. Separate assembly, solve, and complete-total medians are shown; their medians need not sum. Five samples follow a warm-up. Small-grid timings are noise sensitive.

For Neumann data at target 1×10^{-2}, the selected FO/G grids are $N = 128$ and $N = 8$; their measured median cost ratio $T_{\mathrm{FO}}/T_{\mathrm{G}}$ is 1.4×10^2.

For Neumann data at target 5×10^{-3}, the selected FO/G grids are $N = 256$ and $N = 8$; their measured median cost ratio $T_{\mathrm{FO}}/T_{\mathrm{G}}$ is 8.4×10^2.

For Robin data at target 1×10^{-2}, the selected FO/G grids are $N = 64$ and $N = 4$; their measured median cost ratio $T_{\mathrm{FO}}/T_{\mathrm{G}}$ is 37.

For Robin data at target 5×10^{-3}, the selected FO/G grids are $N = 128$ and $N = 8$; their measured median cost ratio $T_{\mathrm{FO}}/T_{\mathrm{G}}$ is 1.4×10^2.

For the successful comparisons on this implementation, G reaches the thresholds on much smaller systems and has lower measured median cost. That is a measured conclusion for these data, candidate grids and machine, rather than a hardware-independent complexity claim. Small-grid timings close to a millisecond are noise-sensitive; ratios are stated to two significant figures. Timings are regenerated by the driver; their ratios should not be transplanted to another machine. No direct/CG runtime crossover is claimed from the single-grid CG cross-check.

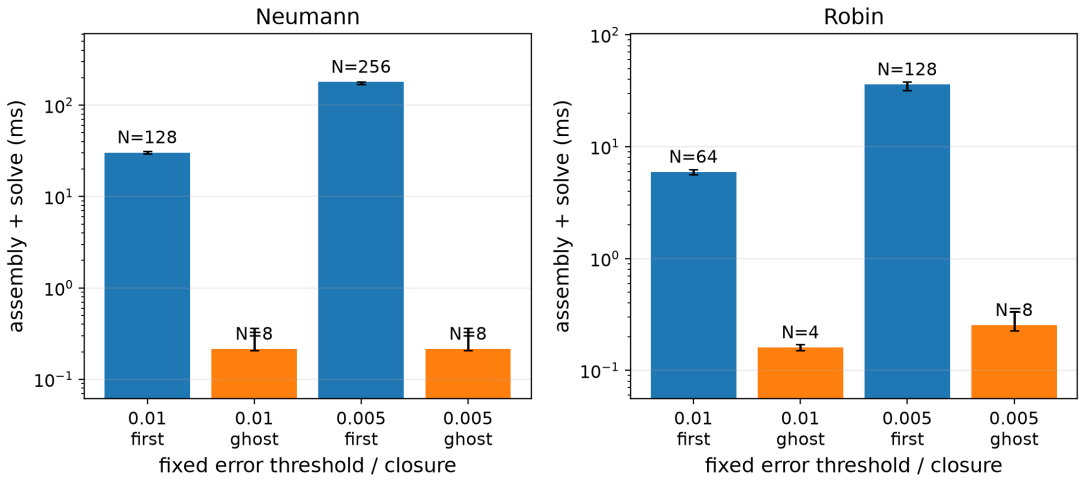


Figure 9: Measured cost under the same error thresholds within each boundary variant. The vertical axis is logarithmic. Error bars show the five-sample minimum–maximum; labels identify the selected grids. Only measured successful candidates are displayed.

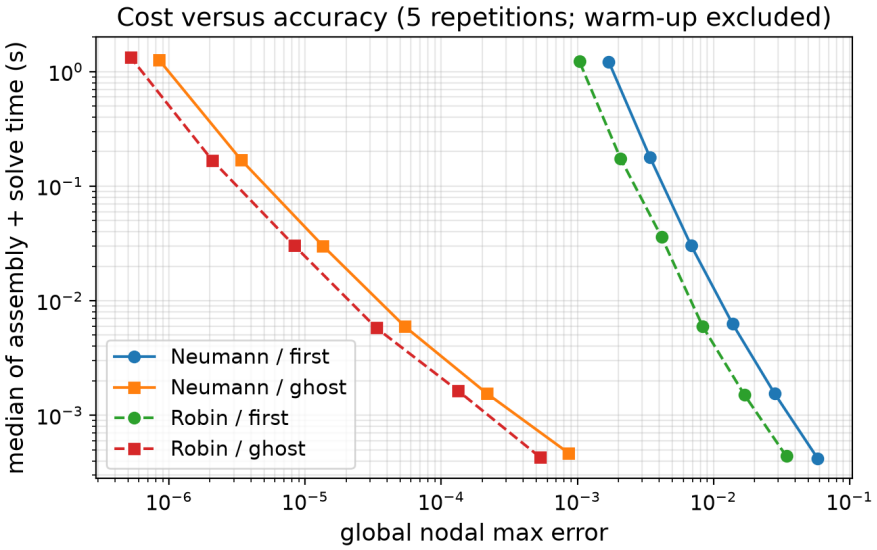


Figure 10: The full error–cost trajectories on the candidate grids $N = 4, \ldots, 512$. Cost is the median of measured assembly-plus-solve totals, including factorisation. This provides context; quantitative fixed-target comparisons use Table 12, not a visual comparison of unmatched points.

7.6 Extension: the pure-Neumann problem

This additional experiment is *not* the mixed problem above. All four edges now carry the outward flux $q = \partial_n e^{x+y}$ and all $(N+1)^2$ vertices, including corners, are unknown. The continuous compatibility condition is $\int_\Omega f + \int_{\partial\Omega} q = 0$. The FO scheme uses one-sided normal derivatives at edge nodes and the sum of the two outward derivative rows at corners. The G scheme uses centred ghost elimination in boundary PDE rows, adding both normal flux contributions at corners. Thus the finite-grid compatibility conditions need not agree.

For each singular matrix we compute a unit-sum left-null vector w and its defect $w^T b$. The constant right-null vector is tested through $\|A\mathbf{1}\|_\infty$. We solve the bordered system

$$\begin{bmatrix} A & \mathbf{1} \\ \mathbf{1}^T & 0 \end{bmatrix} \begin{bmatrix} U \\ \lambda \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix},$$

so U has arithmetic mean zero and $\lambda = w^T b$ is the compatibility correction. Errors compare U with the exact nodal values after the same mean-zero shift. Table 13 and Figure 11 show that the FO defect and gauge error decrease approximately linearly, whereas G decreases approximately quadratically. The independent trapezoidal compatibility quadrature and left-null diagnostics are saved in `data/pure_neumann_depth.csv`. The nonzero finite-grid defect is not concealed by the gauge correction.

Closure	N	$\|A\mathbf{1}\|_\infty$	Compatibility defect	Gauge error	Residual
FO	8	0	1.61×10^{-1}	1.86×10^{-1}	1.2×10^{-13}
FO	16	0	8.01×10^{-2}	9.45×10^{-2}	6.7×10^{-13}
FO	32	0	4.00×10^{-2}	4.76×10^{-2}	8.9×10^{-12}
FO	64	0	2.00×10^{-2}	2.39×10^{-2}	2.9×10^{-11}
G	8	0	-7.70×10^{-3}	1.21×10^{-2}	2.1×10^{-13}
G	16	0	-1.92×10^{-3}	3.05×10^{-3}	6.8×10^{-13}
G	32	0	-4.81×10^{-4}	7.65×10^{-4}	4.4×10^{-12}
G	64	0	-1.20×10^{-4}	1.92×10^{-4}	1.9×10^{-11}

Table 13: Pure-Neumann extension with all boundary vertices unknown. Compatibility uses a unit-sum left null vector; the bordered solve fixes arithmetic mean zero and applies the displayed scalar compatibility correction. This is a distinct depth experiment, not the mixed problem's matrix.

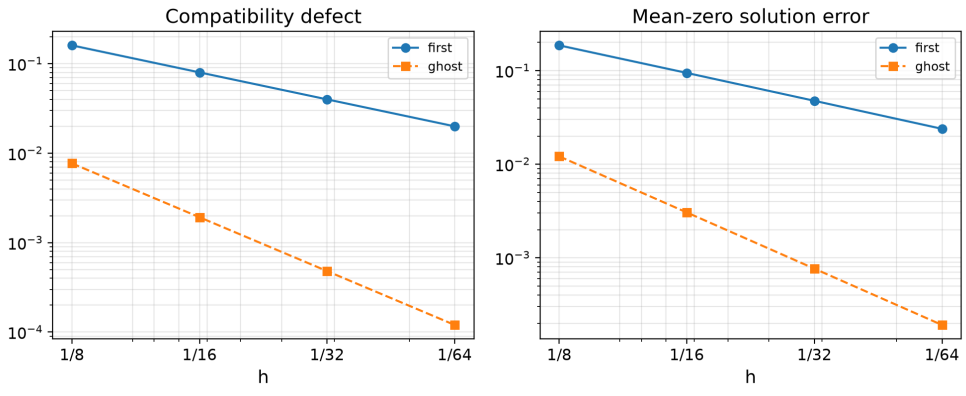


Figure 11: Pure-Neumann finite-grid compatibility defects and mean-zero solution errors for two distinct boundary closures.

8 Conclusions and limitations

The right-edge equation controls the accuracy retained by the mixed Poisson discretisation. The two raw boundary defects are $O(h)$, but the comparison estimate weights the G defect by h. Together with the $O(h^2)$ interior defect, this yields the FO $O(h)$ and G $O(h^2)$ maximum-norm bounds. For e^{x+y}, the separate FO lower bound proves that first order is sharp in the boundary and global maximum norms. The weighted energy identity supplies a second stability route and discrete L^2/energy bounds for the actual boundary and corner layout.

Six-mesh experiments for both Neumann and Robin data reproduce the predicted approximately first- and second-order behaviour. The interior maximum-error fits exhibit visible finite-grid deviations that are reported rather than rounded to the theoretical orders. Independent boundary rows, different manufactured functions and an uneliminated system check test the equations. Direct/GMRES comparisons and weighted CG show solve differences much smaller than the measured discretisation error on the representative grid. An independent P1 FEM solve confirms the mixed weak-form benchmark at shared nodes. A separate pure-Neumann experiment exposes the different finite-grid compatibility defects and the constant nullspace, with a mean-zero gauge. The implemented positive row weights recover the proved symmetric positive definite structure without asserting symmetry of the raw systems.

At fixed nodal-error thresholds, the measured G costs are lower in the successful comparisons because substantially coarser tested grids meet the same bounds. Neumann FO attains the tighter 5×10^{-3} target at $N = 256$ in the extended candidate set. Cost includes fresh factorisation and is measured as complete assembly-plus-solve totals. The finite candidate set, coarsest-successful-grid policy and timing variability limit the scope of this efficiency conclusion.

The mathematical assumptions remain essential: bounded derivatives, fixed nonnegative β, exact imposed Dirichlet values and the stated square geometry. Smoothness is satisfied by the manufactured function and is not proved for arbitrary mixed data or corner-singular problems. The G theorem is an upper bound without a matching lower bound. Floating-point diagnostics and representative solver cross-checks are evidence for the reported computations, not universal forward-error certificates. The energy norms depend on the closure; common nodal errors define the cost targets. Finally, this is verification on known solutions rather than experimental validation of a particular device.

The analysis and computations jointly support the conclusion that the ghost-eliminated formulation preserves second-order accuracy for this smooth benchmark, whereas the backward derivative formulation has a provably first-order maximum error. The distinction follows from the complete boundary formulation and its stability, not from the interior stencil alone.

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A Supplementary numerical data

[TODO: One short paragraph: which CSV files in `data/` feed this appendix, which grids they cover, and that displayed values are rounded while the CSV files keep full precision.]

A.1 Nodal errors on every refinement level

[TODO: Define the quantities used in the tables below (refer to the norms in Section 3).]

A.2 Weighted energy errors and identity checks

[TODO: Energy-norm errors and the edge-sum identity defect per grid.]

A.3 Raw-matrix structure and algebraic residuals

[TODO: Asymmetry measures, smallest eigenvalue estimates, componentwise and normwise residuals.]

B Pre-computation predictions and reconciliation

[TODO: Two or three sentences: this appendix records predictions written *before* the corresponding computations, the observed output, and the reconciliation decision. State which predictions come from analysis (theorems, counts, limiting cases) and that no prediction was edited after seeing the output. Items first checked after a run must be labelled as such.]

#	Prediction (before running)	Output (after)	Reconciliation decision
1	**[TODO: e.g. sign pattern / symmetry of the raw and weighted matrix]**		
2	**[TODO: expected order of FO and G, and the norm]**		
3	**[TODO: nonzero count from the sparsity formula]**		
4	**[TODO: where the maximum error is located]**		
5	**[TODO: limiting case with a known answer]**		
6	**[TODO: what the solver tolerance must be below]**		

Table 14: Predictions, outputs and reconciliation decisions. Each disagreement is classified as a defect in the model, the code, or the prediction; no prediction was altered to match an output.

[TODO: Short discussion of any disagreement and how it was resolved.]

C AI Transparency Log

C.1 Supplementary five-gate audit record

[TODO: Identify the AI-generated code subject to audit (file, function, commit) and the date of the audit.]

Gate	Evidence and findings	Outcome
1. Transcription	**[TODO: code compared with the written formulas]**	**[TODO: pass/fail]**
2. Structure and indexing	**[TODO: index map, corner ownership, row layout]**	**[TODO: pass/fail]**
3. Linearisation	**[TODO: state applicability; the problem is linear]**	**[TODO: n/a or result]**
4. Independent reproduction	**[TODO: hand rows, uneliminated system, FEM oracle]**	**[TODO: pass/fail]**
5. Interrogation	**[TODO: questions put to the code and the answers]**	**[TODO: pass/fail]**

Table 15: Five-gate audit of the AI-generated solver code.

D Individual Contribution Statement